



STAT 254

Lecture Notes

Department of Mathematics and Statistics
Camosun College

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Set 1 Population and Sample

A population is the entire group of individuals about which we wish to study.

Example: Some examples of populations:

1. All I-beams being made by a particular manufacturer.
2. All current Camosun students.
3. All Canadians who will be eligible to vote in an upcoming election.
4. All people who will at some point take a particular blood pressure medication.

A sample is any subset of the population.



Example: We might select every tenth I-beam made in the manufacturing process, or take a random sample of 500 voters, or select 40 patients with high blood pressure from a local hospital.

⇒ We are usually interested in particular characteristics of the objects in the population. A variable is any characteristic that may vary from one member of the population to another.

The data that results from our observations may consist of one variable (a **univariate** data set), two variables (a **bivariate** data set), or many variables (a **multivariate** data set).

Descriptive Statistics and Inferential Statistics:

In the branch of **descriptive statistics**, we organize, summarize, display, and describe features of the data.

Example: Some examples of descriptive statistics:

1. What is the greatest tensile strength recorded? What is the range of recorded tensile strengths?
2. What is the average of recorded tensile strengths?
3. What proportion of the sample of voters is older than 65?
4. What is the average weight of the sample of people taking blood pressure medication? How spread out are the measurements for resting heart rate?

In the branch of **inferential statistics**, we try to discover some property or general pattern about a large group (population) by studying a smaller group of people (sample) in the hopes that the results will generalize to the larger group. In another words, we try to draw conclusions about the population based on the measurements from the sample.

Example: If we ask residents of New York City their opinion about their mayor. We would probably poll a few thousand individuals in New York City in an attempt to find out how the city as a whole views their mayor.

Example: Some sorts of questions inferential statistics answers:

1. Based on our sample of I-beams, what is a likely range of values of tensile strengths for all I-beams made by the manufacturer?
2. Based on our survey, which party is likely to win the election?
3. Based on our sample of patients, can we conclude that there a relationship between weight and blood pressure?
4. Based on our sample of 40 Camosun students, is there reason to believe that the average time to complete an undergraduate degree is longer than 4.5 years?

parameter $\leftarrow$ popn. ; statistic $\leftarrow$ sample

A **parameter** is some sort of numerical summary of a population. A **statistic** is some sort of numerical summary of a sample.

Example: If we are interested in the population of all current Camosun students, then the highest grade-point average of all students would be an example of a parameter. If I were to take a sample of 40 Camosun students, then the highest grade-point average of these 40 students would be a statistic.

Note: Usually, the value of a parameter is unknown (and unknowable). One of our goals with inferential statistics is to try and draw conclusions about parameters a related statistic.

Determine whether the underlined words refer to a:

(a) Population

(b) Sample

(c) Parameter

(d) Statistic

1. We wish to study poplar trees, so we make a selection of 15 poplar trees in a forest. *sample.*

2. From our selection of 15 poplar trees, the tallest tree measures 1.9 meters in height. *statistic*

3. We wish to study the effects of a new medication on people with bile duct cancer. *pop'n.*

4. It is known that for people with bile duct cancer, the average survival time after diagnosis is 1.5 years. *parameter.*

5. A newspaper wants to determine the feelings of Victoria residents regarding a bridge to the mainland. *pop'n.*

6. The newspaper phones 500 Victoria residents. *sample*

7. It is found that 95% of these people are in favour of a bridge. *statistic.*

Set 2 Measures of location and variability

The sample size is the number of observations. For a single sample, we denote the sample size by n .

(If we have multiple samples, we use subscripts. For example, n_1 is the size of the first sample, n_2 is the size of the second sample, and so on.)

Individual observations for a single sample will be denoted by $x_1, x_2, \dots, x_n$.

Note: We should not assume that the observations are sorted from smallest to largest.

Measures of Central Tendency: A measure of central tendency is a single value that attempts to describe a set of data by identifying the central position within that set of data.

Mean, Median, and Mode:

Mean

"x bar" The **mean** is another word for the **average** of a set of observations.

$\bar{x}$ is the sample mean (mean of all observations from a sample). *Statistic*

"mu" μ is the populations mean (mean of all observations from a population). *parameter*

Usually, μ is unknown (and unknowable). We use $\bar{x}$ as an estimate for μ .

Formula:

$$\bar{x} = \frac{x_1 + x_2 + \dots + x_n}{n} = \frac{\sum_{i=1}^n x_i}{n}$$

Example: Find the mean of the following sample.

10, 6, 12, 7, 3, 6, 7

$$\bar{x} = \frac{10 + 6 + 12 + \dots + 7}{7}$$

Median

$\text{median} = \begin{cases} \text{single middle value} & \text{if } n \text{ is odd.} \\ \text{average of two middle values} & \text{if } n \text{ is even} \end{cases}$

The **median** is the "middle" observation of an ordered data set. We use $\tilde{x}$ to represent the sample median, and use $\tilde{\mu}$ to represent the population median.

If we have an odd number of observations, the median is the middle observation. If we have an even number of observations, the median is the average of the two middle observations.

Example: Find the median of 6, 9, 19, 11, 3.

3, 6, 9, 11, 19 $n=5$
↓
median.

Example: Find the median of 3, 8, 11, 14, 19, 20.

3, 8, 11, 14, 19, 20 $n=6$
 $\text{median} = \frac{11+14}{2} = 12.5$

Mode

The **mode** is the observation that occurs the most number of times. The mode of 3, 5, 9, 9, 9, 5 is 9, since 9 occurs the most often. There might be one mode, many modes, or no modes in a set of observations.

The median is unaffected by outliers (extreme observations), while the mean is affected by outliers.

Example: Consider the two samples: 3, 4, 5, 6, 7 and 3, 4, 5, 6, 100.

$$\text{median : } \tilde{x}_1 = 5, \quad \tilde{x}_2 = 5$$

$$\text{mean : } \bar{x}_1 = 5, \quad \bar{x}_2 = \frac{3+4+5+6+100}{5} = 23.6$$

A trimmed mean is a compromise between the median and the mean. A $k\%$ trimmed mean is found by eliminating the smallest $k\%$ and largest $k\%$ of the observations, then finding the mean of the remaining observations.

Note that the greater the percentage we trim, the closer the trimmed mean will be to the median.

Data set 1: $\{-1, 0, 1\}$

Data set 2: $\{-1000, 0, 1000\}$

$$\left\{ \begin{array}{l} \text{mean} = 0 \\ \text{median} = 0 \end{array} \right.$$

Variability:

Variability describes how far apart data points lie from each other and from the centre of a distribution. Along with measures of central tendency, measures of variability give you descriptive statistics that summarize your data.

Variability is also referred to as spread, scatter or dispersion. It is most commonly measured with the following:

- (1) **Range**: the difference between the highest and lowest values.
- (2) **Interquartile range**: the range of the middle half of a distribution.
- (3) **Standard deviation**: average distance from the mean.
- (4) **Variance**: average of squared distances from the mean

s^2 and σ^2 denote the sample variance and the population variance, respectively. (Sigma²)

The i^{th} deviation is the difference between x_i and $\bar{x}$, $x_i - \bar{x}$.

A useful fact about deviations:

$$\sum (x_i - \bar{x}) = 0$$

$$\left\{ \begin{array}{l} \bar{x} = \frac{\sum x_i}{n} \\ n\bar{x} = \sum x_i \\ \sum x_i - n\bar{x} = 0 \\ \sum x_i - \sum \bar{x} = 0 \\ \sum (x_i - \bar{x}) = 0 \end{array} \right.$$

The sample variance, denoted s^2 is:

$$\underline{s^2} = \frac{\sum (x_i - \bar{x})^2}{n - 1}$$

where n is sample size.

$s^2 \rightarrow \sigma^2$
estimate
 $\downarrow$
 $(n-1)$ gives us an unbiased estimate
not "n"

and the **population variance**, denoted σ^2 is:

$$\sigma^2 = \frac{\sum (x_i - \mu)^2}{\underline{N}}$$

where N is the population size.

s and σ are the sample and population **standard deviation**, respectively.

Standard deviation is the square root of the variance.

$$s = \sqrt{s^2} ; \sigma = \sqrt{\sigma^2}$$

Example: Find the variance and standard deviation of the following sample:

7, 7, 9, 15, 16, 17, 19, 21, 22, 40

$$n = 10, \bar{x} = 17.3$$

$$\begin{aligned} s^2 &= \frac{1}{n-1} \sum (x_i - \bar{x})^2 = \frac{1}{10-1} [(7-17.3)^2 + (7-17.3)^2 + \dots \\ &\quad + (40-17.3)^2] \\ &= 93.567 \end{aligned}$$

$$s = \sqrt{s^2} = 9.6730$$

$$\bar{x} = \frac{\sum x_i}{n} \Rightarrow n\bar{x} = \sum x_i$$

A **computational formula** exists to assist with hand calculation of sample variance. It uses the fact that

$$\sum (x_i - \bar{x})^2 = \sum (x_i^2) - n\bar{x}^2$$

From this, we have:

$$s^2 = \frac{\sum (x_i^2) - n\bar{x}^2}{n - 1}$$

$$\begin{cases} \sum (x_i - \bar{x})^2 = \sum (x_i^2 - 2\bar{x}x_i + \bar{x}^2) \\ = \sum x_i^2 - \sum 2\bar{x} \cdot x_i + \sum \bar{x}^2 \\ = \sum x_i^2 - 2\bar{x} \sum x_i + n\bar{x}^2 \\ = \sum x_i^2 - 2n\bar{x}^2 + n\bar{x}^2 \\ = \sum x_i^2 - n\bar{x}^2 \end{cases}$$

Example: In the example we just did, find the variance and standard deviation by using the computational formula:

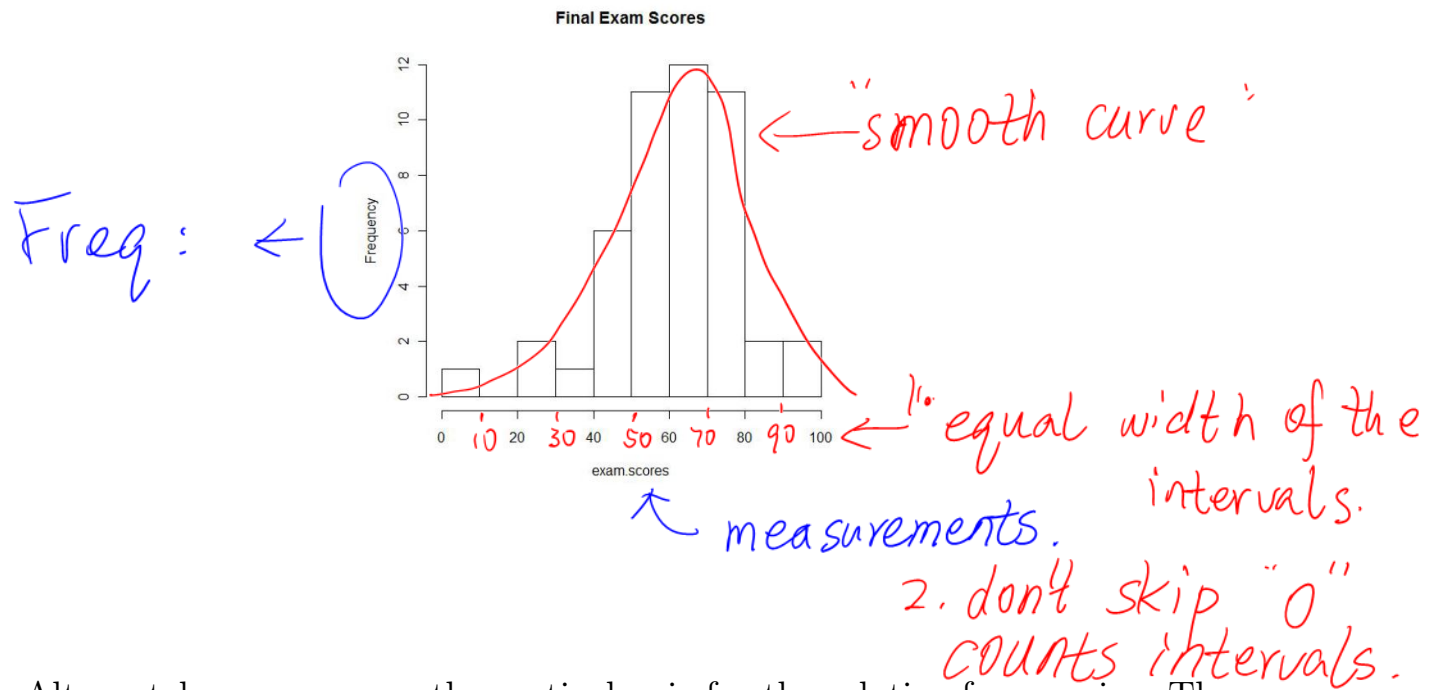
$$n = 10, \bar{x} = 17.3$$

$$s^2 = \frac{1}{10 - 1} [7^2 + 7^2 + \dots + 40^2 - 10 \cdot 17.3^2]$$

$$= 93.567$$

$$s = \sqrt{s^2} = 9.6730.$$

A **histogram** is a visual representation of a frequency distribution (a table counting the number of observations per category/interval). The horizontal axis is for our measurements, and the vertical axis is for the frequencies. The following is a histogram of 48 final exam scores.



Alternately, we may use the vertical axis for the relative frequencies. The relative frequency is the proportion of times the value occurs.

← proportions.

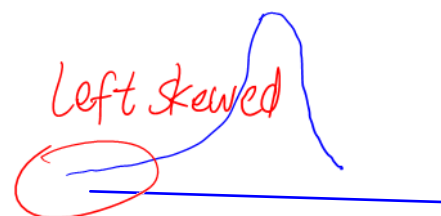
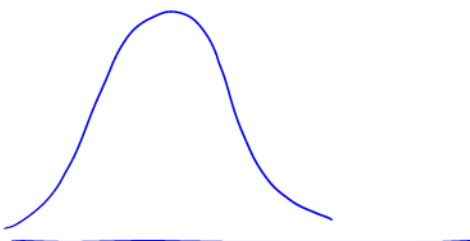
We can imagine our histogram as being approximated by a smooth curve. We can classify our frequency distribution by the shape of these curves.

Curves may be **symmetric**. They may have one peak (**unimodal**), two peaks (**bimodal**), or many peaks (**multimodal**). For **unimodal distributions**, they may be **positively skewed** (the right tail is "stretched") or **negatively skewed** (the left tail is "stretched").

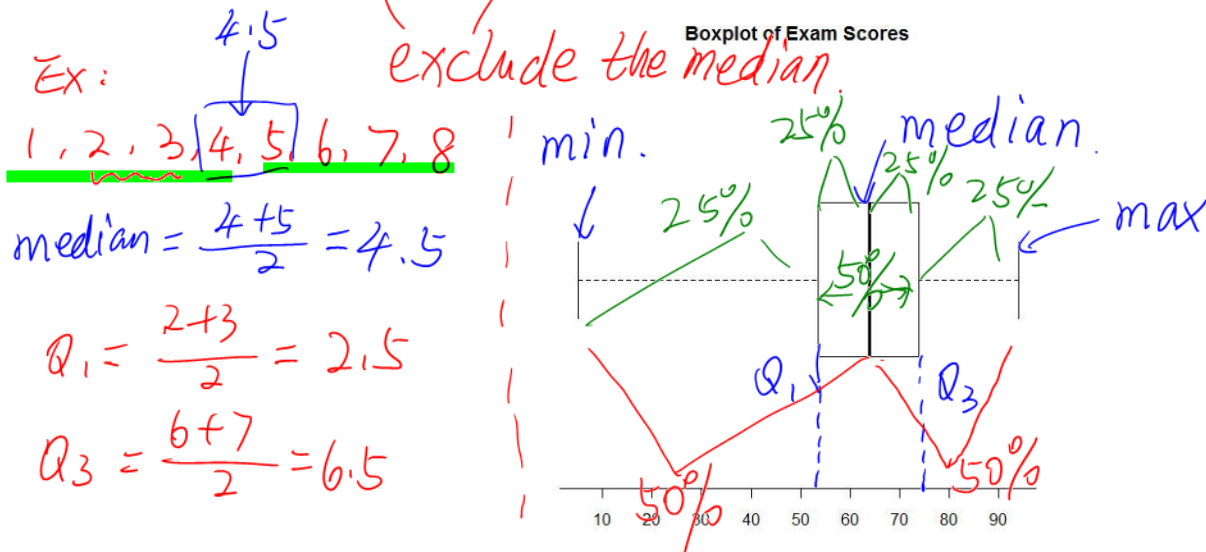
1. symmetric.

2. positively skewed

3. negatively skewed



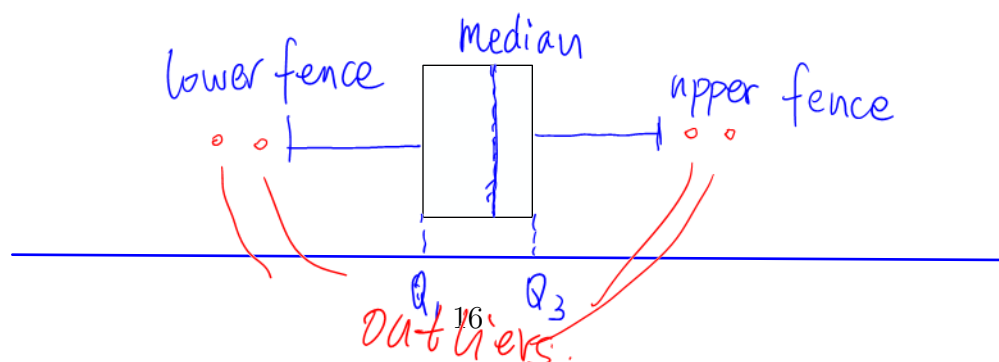
Another useful visual summary is a **boxplot**. The boxplot allows the viewer to visually identify the median, the lower quartile, Q_1 (the median of the lower half of the observations), and the upper quartile Q_3 (the median of the upper half of the observations).



Modified boxplot: Interquartile range (IQR) = $Q_3 - Q_1$ and IQR Rule.

IQR Rule: Lower fence = $Q_1 - 1.5 \cdot \text{IQR}$
 upper fence = $Q_3 + 1.5 \cdot \text{IQR}$.

any points outside of lower and upper fence interval
 $\hookrightarrow$ outliers.



Examples: Suppose we are summarizing a data set. When might we choose to use the median rather than the mean as a measure of the centre?

- ~~(a)~~ When you suspect there are no outliers in the data set.
- ~~(b)~~ Whenever you feel like it either is appropriate to use at any time.
- ~~(c)~~ When you suspect there are many outliers in the data set.

Suppose we have two data sets. The first data set has a standard deviation of 5 and the second data set has a standard deviation of 10.

$$S_1 = 5$$

$$S_2 = 10$$

- ~~(a)~~ The first data set is more "spread out" than the second data set.
- ~~(b)~~ The second data set is more "spread out" than the first data set.
- ~~(c)~~ We don't have enough information to tell which data set is more "spread out".

Calculator STAT Mode: We wish to investigate the number of defective items that a machine in a factory makes. We watch the machine for eight days, and record the number of defective items made each day.:

0, 2, 3, 1, 5, 0, 0, 2

Find the sample mean and variance.

$$\bar{x} = 1.625$$

$$S = 1.767766953$$

$$S^2 = 3.125$$

Set 3 Sample Spaces and Events

Experiment: any activity or process whose outcome is subject to uncertainty.

Example: Flipping a coin three times and noticing the sequence of heads and tails is an experiment.

$$\{HHH, HHT, HTH, THH, TTH, THT, HTT, TTT\} = S$$

The results are the **outcomes** of the experiment.

The set of all possible outcomes is called the **sample space**. We denote the sample space by S . The sample space may contain a finite or an infinite number of outcomes. A single outcome is sometimes called a **sample point**.

Any subset, or collection of outcomes, is called an **event**. If an event consists of a single outcome, we call it a **simple event**. Otherwise, we call it a **compound event**.

Example: Consider the experiment where we flip a coin three times and note the sequence of heads and tails. What is the sample space of this experiment?

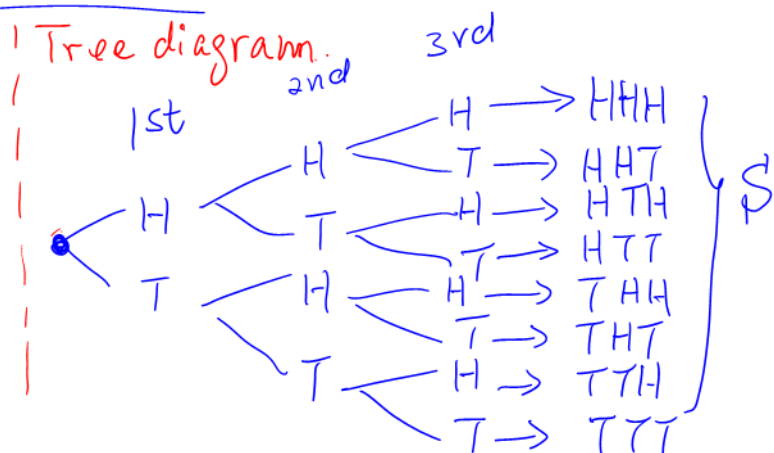
Show the events:

(1): two heads. (2): three heads. (3): at least two heads.

let $A = \text{two heads} = \{HHT, HTH, THH\}$

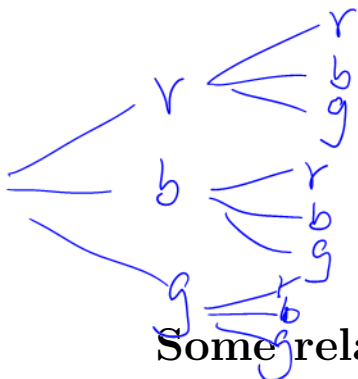
let $B = \text{three heads} = \{HHH\}$

let $C = \text{at least two heads} = \{HHH, HHT, THH, HTH\}$



Tree Diagram and Sample Space:

Example: Suppose that a box contains ^r red, ^b blue, and ^g green marbles (several of each colour). Two marbles are selected one at a time from the box, and the sequence of colours is noted. Use the tree diagram to find the sample space?

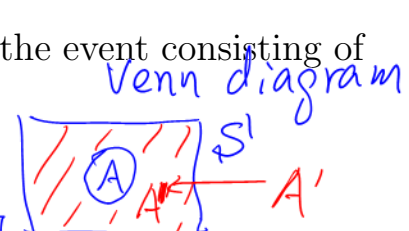


$$S' = \{ rr, rb, rg, br, bb, bg, gr, gb, gg \}$$

Some relations from Set Theory

We say that an event occurs if one of its sample points is an observed when we carry out the experiment:

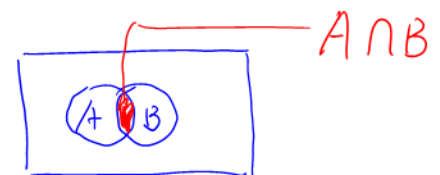
- (a) The complement of an event A, denoted A' , is the event consisting of all outcomes for which A does not occur.



- (b) The union of events A and B, denoted $A \cup B$ and read "A or B", is the event consisting of all outcomes for which A or B (or both) occur.



- (c) The intersection of events A and B, denoted by $A \cap B$ and read "A and B", is the event consisting of all outcomes for which A and B both occur.



Venn diagrams can be used to show whether or not there is overlap between two or more sets.

Note: In general, the word "or" is used in the inclusive sense in mathematics (i.e. this or that or both).

Example: Return to the three events from our coin flipping experiment.

$$A = \{HTH, HHT, THT\}$$

$$B = \{HHH\}$$

$$C = \{HHH, HHT, HTH, THT\}$$

$$A \cup B = \{HTH, HHT, THT, HHH\}$$

$$B \cup C = \{HHH, HHT, HTH, THT\} \quad \left\{ \begin{array}{l} B \cap C = \{HHH\} \\ A \cap B = \text{empty} = \emptyset \text{ "phi"} \end{array} \right.$$

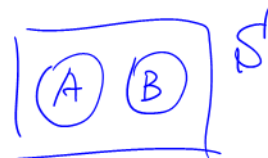
$$A \cap B = \text{empty} = \emptyset \text{ "phi"}$$

We call the event S the **guaranteed or certain event**, because in any trial of an experiment, one of its sample points will be observed.

The event which consists of no outcomes is denoted by $\emptyset$. We call empty set ; the **impossible event** or **null event**, because it can never occur.

If for events A and B, we have $A \cap B = \emptyset$, then we say that A and B are disjoint or mutually exclusive events.

M.E.



A and B are M.E.

Set 4 Probability

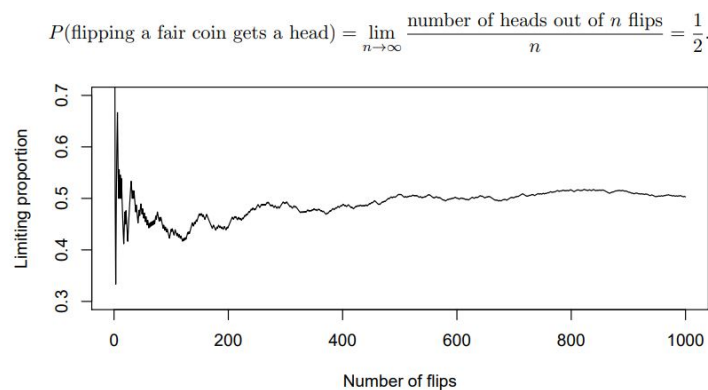
Probability is used to quantify the likelihood, or chance, that an outcome of a random experiment will occur. The probability of an event E is denoted by $P(E)$.

Probability:

One approach to calculating probability within the frequency interpretation utilizes a conceptual framework involving the repeated execution of the random experiment.

Example: we want to quantify the probability of the event that flipping a fair coin gets a head. One way is to flip a fair coin n (must be a large number) times, and record how many times you get a head.

We have:



This type of experiment is said of equally likely outcomes.

Equally Likely Outcomes: Whenever a sample space consists of N possible outcomes that are equally likely, the probability of each outcome is $1/N$.

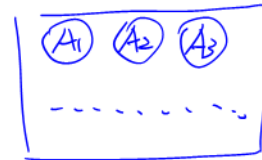
The other approach, classical (theoretical) approach: this approach figures out probability using math and special counting tricks, instead of imagining lots of repeated trials.

Example: Consider drawing a single card from a standard deck of 52 cards. We want to find the probability of drawing an Ace. Using the classical probability formula:

$$P(\text{drawing an Ace}) = 4/52 = 1/13$$

Probability Axioms:

- ① • $P(S) = 1$.
- ② • $0 \leq P(A) \leq 1$ for any event A
- ③ • If $A_1, A_2, A_3, \dots$ is an infinite collection of disjoint events, then



$$P(A_1 \cup A_2 \cup A_3 \cup \dots) = \sum P(A_i)$$

From these axioms, we can derive other properties of probability, including:

- ④ • $P(\emptyset) = 0$ $S' = \emptyset$; S and $\emptyset$ are m.e.

- ⑤ • If $A_1, A_2, A_3, \dots, A_k$ is an collection of k disjoint events

$$P(A_1 \cup A_2 \cup A_3 \cup \dots \cup A_k) = \sum P(A_i)$$

- ⑥ • $P(A) = 1 - P(A')$ for any event A .

A and A' are m.e.

eg. $P(A \cup B) = P(A) + P(B)$
 $\downarrow$
 A and B are m.e.

$$P(A \cup A') = P(A) + P(A') = 1$$

rearrange : $P(A \cap B) = P(A) + P(B) - P(A \cup B)$

Addition rule: For any events A and B.

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Example: Suppose that 12% of Camosun students are taking a Statistics class, 28% are taking a Chemistry class, and that 35% are taking a Statistics or Chemistry class.

If we select a Camosun student at random, what is the probability that they are taking both a Statistics class and a Chemistry class?

$$P(S) = 0.12 \quad P(C) = 0.28 \quad P(S \cup C) = 0.35$$

$$P(S \cap C) = ?$$

$$P(S \cap C) = P(S) + P(C) - P(S \cup C) = 0.12 + 0.28 - 0.35 = 0.05$$

Example: A particular item may be rated as "Defective", "Satisfactory", or "Excellent". Suppose that 38% of all items are rated Excellent and 53% are rated Satisfactory. What proportion of items are rated Defective?

$$P(E) = 0.38 \quad P(S) = 0.53 \quad P(D) = ?$$

E, S, and D are M.E.



$$\begin{aligned} P(D) &= 1 - P(E) - P(S) \\ &= 1 - 0.38 - 0.53 \\ &= 0.09 \end{aligned}$$

Set 5 Conditional Probability

In some cases, we may perform an experiment, and already know something about the outcome.

Example: Suppose I have a bag containing 5 red marbles and 5 green marbles. I will ^{select} two marbles from the bag, one at a time, without replacement.

- If the first marble is red, then the probability that the second marble will be red will be $4/9$.

$$P(\text{2nd Red if 1st red}) = 4/9$$

condition

- If the first marble is green, then the probability that the second marble will be red will be $5/9$.

$$P(\text{2nd Red if 1st Green}) = 5/9$$

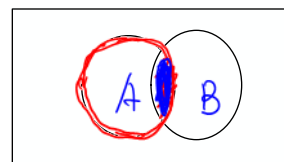
condition

These probabilities are known as **conditional probabilities**. They are the probability of an event occurring, taking into account some information about the experiment (the **condition** or **conditioning event**).

Notation: $P(B|A)$ denotes the probability that B will occur given that A occurs.

given

$$P(B|A) = \frac{P(B \cap A)}{P(A)}$$



Example: Consider the results of the following survey of 250 single-crop farms:

	^W Wheat	^C Corn	^{Soy} Soy
^A Alberta	69	15	16 $\Rightarrow 100$
^S Saskatchewan	61	65	24 $\Rightarrow 150$

$n=250$

Handwritten notes: "A ∩ Soy", "Conditional Prob. Given, Suppose--"

Suppose that a single-crop farm is selected at random. If the farm is in Alberta, what is the probability the farm grows soy?

$$P(\text{Soy} | A) = \frac{P(\text{Soy} \cap A)}{P(A)} = \frac{16/250}{100/250} = \frac{16}{100}$$

Example: If a farm which grows soy is selected, what is the probability that the farm is in Alberta?

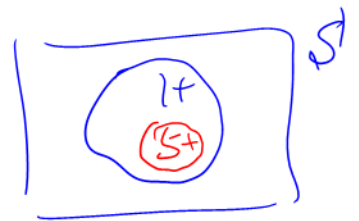
$$P(A | \text{Soy}) = \frac{P(A \cap \text{Soy})}{P(\text{Soy})} = \frac{16/250}{40/250} = \frac{16}{40}$$

Example: Suppose 80% of all Canadians exercise one or more days a week,
and also, that 20% of all Canadians exercise at five or more days a week.

If we randomly select a Canadian who exercises at least one day a week,
what is the probability that this Canadian exercises five or more days a week?

$$P(1^+) = 0.8 \quad P(5^+) = 0.2$$

$$P(5^+ | 1^+) = \frac{P(5^+ \cap 1^+)}{P(1^+)}$$



$$= \frac{P(5^+)}{P(1^+)}$$

$$= \frac{0.2}{0.8} = 0.25$$

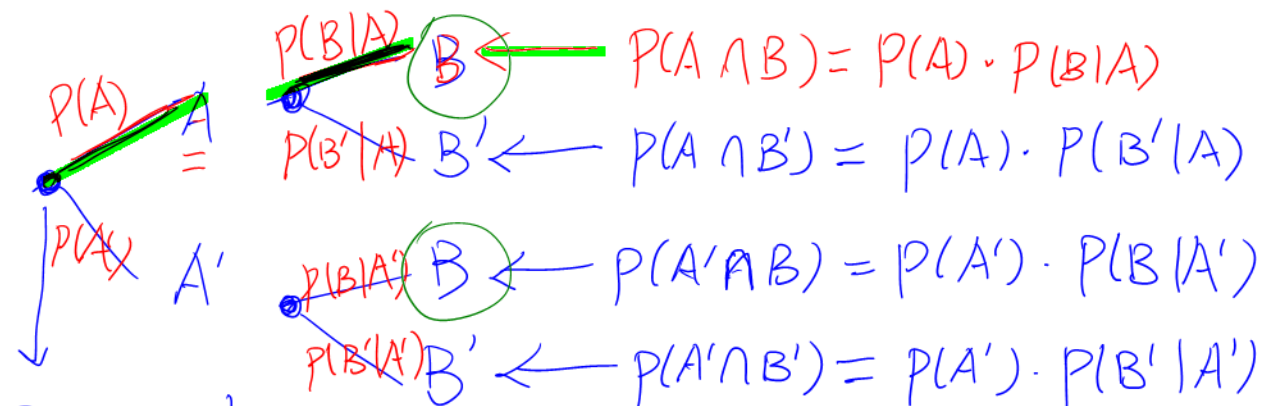
$$P(B|A) = \frac{P(B \cap A)}{P(A)}$$

The Multiplication Rule: Rearrange the equation for condition probability to get the following:

$$P(B \cap A) = P(B|A)P(A)$$

This form of the conditional probability formula is useful when using a **tree diagram** to model an experiment.

two events : A , B. and $P(B|A)$



①. events must be M.E.

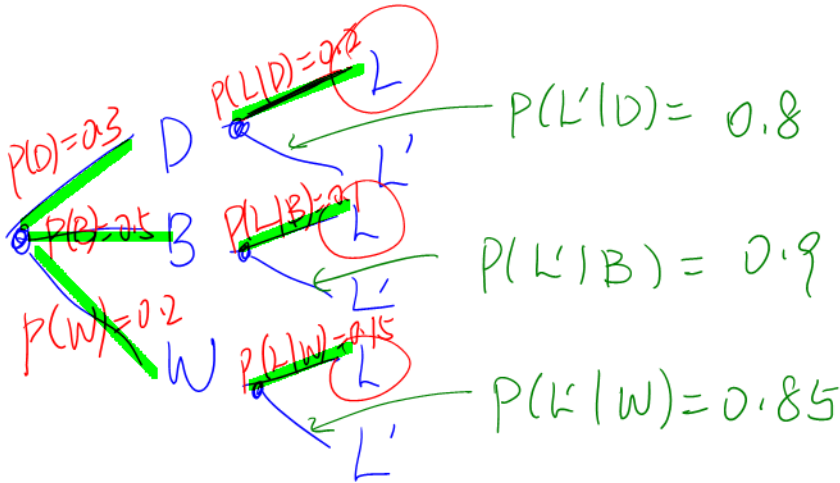
② the sum of the probs. of all events is = 1.

$$P(B) = P(A) \cdot P(B|A) + P(A') \cdot P(B|A')$$

Example: Suppose that 30% of all students drive to school, 50% take the bus, and 20% walk. Of those who drive, 20% are usually late for their first class of the day. Of those who take the bus, 10% are usually late for their first class of the day. Of those who walk, 15% are usually late for their first class of the day. What is the probability that a randomly selected student is regularly late for their first class?

$$P(D) = 0.3 \quad P(B) = 0.5 \quad P(W) = 0.2$$

$$P(L|D) = 0.2 \quad P(L|B) = 0.1 \quad P(L|W) = 0.15$$



$$P(L) = 0.3 \cdot 0.2 + 0.5 \cdot 0.1 + 0.2 \cdot 0.15$$

$$= 0.14$$

Set 6 Bayes' Theorem



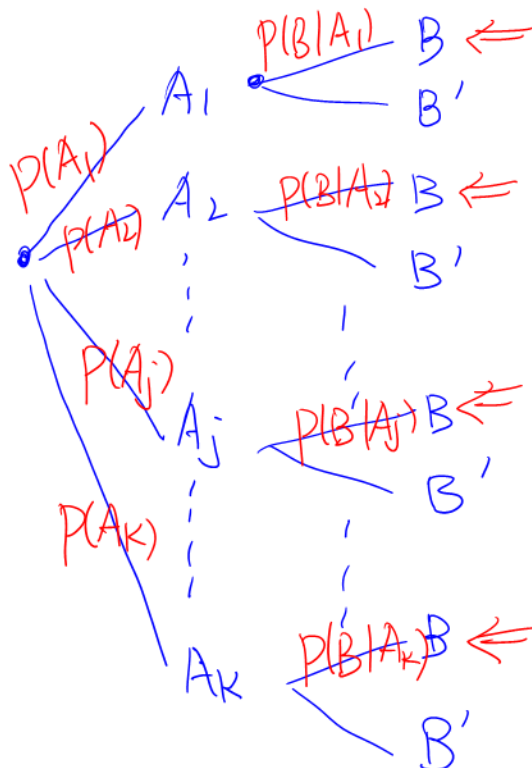
Exhaustive: a collection of events $A_1, A_2, \dots, A_k$ are exhaustive if they completely cover the sample space. That is, if these events are exhaustive, then $A_1 \cup A_2 \cup A_3 \dots \cup A_k = S$.

Law of Total Probability: If $A_1, A_2, \dots, A_k$ are a collection of mutually exclusive and exhaustive events (i.e. the events $A_1, A_2, \dots, A_k$ partition the sample space), then for any event B we have:



$$P(B) = P(B \cap A_1) + P(B \cap A_2) + \dots P(B \cap A_k)$$

$$= P(B|A_1)P(A_1) + P(B|A_2)P(A_2) + \dots P(B|A_k)P(A_k)$$



$$P(B) = P(A_1) \cdot P(B|A_1) + P(A_2) \cdot P(B|A_2) + \dots + P(A_k) \cdot P(B|A_k)$$

Bayes' Theorem: If $A_1, A_2, \dots, A_k$ partition the sample space S , then for any event B (where $P(B) \neq 0$) we have the following, for $1 \leq i \leq k$:

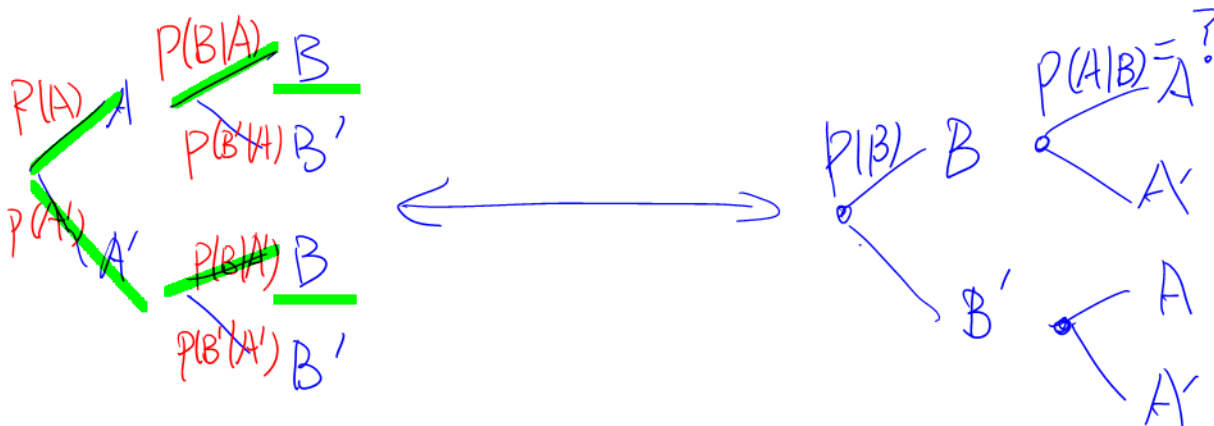
$$P(A_i|B) = \frac{P(A_i \cap B)}{P(B)}$$

$$= \frac{P(B|A_i)P(A_i)}{P(B|A_1)P(A_1) + P(B|A_2)P(A_2) + \dots P(B|A_k)P(A_k)}$$

Note: This theorem is used when we know something about B , the end of an experiment, and wish to know something about how an experiment began.

Tree diagrams: the branches partition the sample space; if we've drawn a tree diagram, then we don't need to formally make use of this theorem.

Give $P(B|A)$ Bayes Thm. $\rightarrow$ Find $P(A|B) = ?$



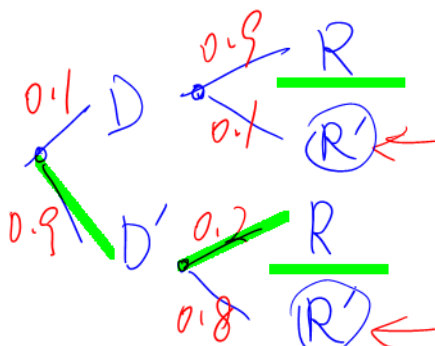
$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{P(A) \cdot P(B|A)}{P(A) \cdot P(B|A) + P(A') \cdot P(B|A')}$$

Bayes' Thm.

Example: The probability of an item on a certain production line being defective is 0.1. If an item is defective, the probability that the inspector will remove it from the line is 0.9. If an item is not defective, the probability that the inspector will remove it from the line is 0.2.

If an item is removed from the production line, what is the probability that it isn't actually defective?

$$P(D) = 0.1 \quad P(R|D) = 0.9 \quad ; \quad P(R|D') = 0.2$$



$$P(D'|R) = \frac{P(D' \cap R)}{P(R)} = \frac{0.9 \cdot 0.2}{0.1 \cdot 0.9 + 0.9 \cdot 0.2} = 0.667$$

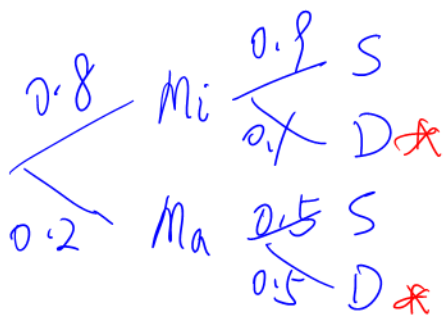
Example: If an item is not removed by the inspector, what is the probability that it is actually defective?

$$P(D|R') = \frac{P(D \cap R')}{P(R')} = \frac{0.1 \cdot 0.1}{0.1 \cdot 0.1 + 0.9 \cdot 0.8} = 0.0137$$

Example: At a factory, 80% of all items produced are judged to need minor repairs, and 20% need major repairs. Of the items needing minor repairs, 90% are eventually Sold, and the remainder are Destroyed. Of the items needing major repairs, 50% eventually Sold, and the remainder are Destroyed.

What is the probability that a randomly selected item will be destroyed?

$$P(M_i) = 0.8 \quad P(M_a) = 0.2 \quad ; \quad P(S|M_i) = 0.9 \quad ; \quad P(S|M_a) = 0.5$$



$$P(D) = 0.8 \cdot 0.1 + 0.2 \cdot 0.5 = 0.18$$

Example: If a randomly selected item was destroyed, what is the probability it needed major repairs?

$$P(M_a|D) = \frac{P(M_a \cap D)}{P(D)} = \frac{0.2 \cdot 0.5}{0.18} = 0.556$$

Set 7 Independence

In some cases, knowing the outcome of one event does not provide any information for a second event, or affect the probability of the second event occurring.

Example: Suppose we will ~~to~~ roll two dice, and were concerned with the events A, that the first roll is a five, and B, the second roll is a four. Suppose I know that A has occurred. This doesn't in any way change the likelihood that B will occur. Since the occurrences of A and B do not affect each other, we say that A and B are **independent events**.

More formally, let A and B be events with nonzero probabilities. We say that A and B are independent events if and only if:

- $P(B|A) = P(B)$
- $P(A|B) = P(A)$
- $P(A \cap B) = P(A)P(B)$

$$P(B|A) \stackrel{\text{indep.}}{=} P(B)$$

||

$$\frac{P(B \cap A)}{P(A)} \stackrel{\text{indep.}}{=} P(B) \Rightarrow P(A \cap B) \stackrel{\text{indep.}}{=} P(A) \cdot P(B)$$

It can be shown that these three statements are equivalent (i.e. if one is true, then all three are true).

Note: if A and B are independent events, then A and B' are independent, A' and B are independent, and also A' and B' are independent.

$$A \text{ and } B \text{ indep.} \iff A \text{ and } B' \text{ indep.} \iff A' \text{ and } B \text{ indep.}$$

$$\updownarrow \\ A' \text{ and } B' \text{ indep.}$$

Example: In a small tree stand, 8% of all trees suffer from root ^Adisease, and 5% of all trees suffer from insect ^Binfestation. Also, it is known that 12% of trees suffer from at least one of these conditions.

Suppose ^AA is the event that a randomly selected tree has root disease, and ^BB is the event that a randomly selected tree has insect infestation. Are A and B independent events?

$$P(A) = 0.08 \quad P(B) = 0.05 \quad ; \quad P(A \cup B) = 0.12$$

$$P(A \cap B) \stackrel{\text{indep.}}{\neq} P(A) \cdot P(B)$$

additional rule : for any two events A and B :
 $P(A \cap B) = P(A) + P(B) - P(A \cup B)$

$$P(A \cap B) = 0.08 + 0.05 - 0.12 = 0.01$$

$$\neq P(A) \cdot P(B) = 0.08 \cdot 0.05 = 0.0040$$

A and B are not independent.

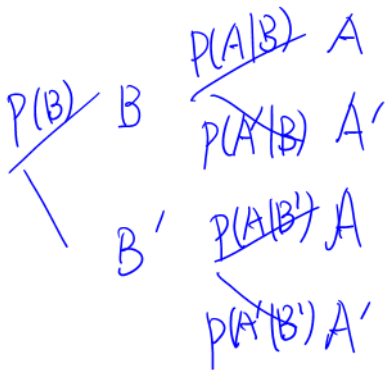
Independence of more than two events:

A collection of events are independent if for every subset of this collection, $A_1, A_2, \dots, A_k$, we have

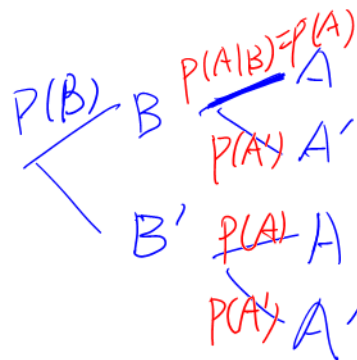
$$\underline{P(A_1 \cap A_2 \cap \dots \cap A_k) = P(A_1)P(A_2)\dots P(A_k)}$$

Much like conditional events, we can use tree diagrams to model problems where the events are independent. We find the solution with a tree diagram in the exact same way as we did for conditional events.

Conditional Prob.

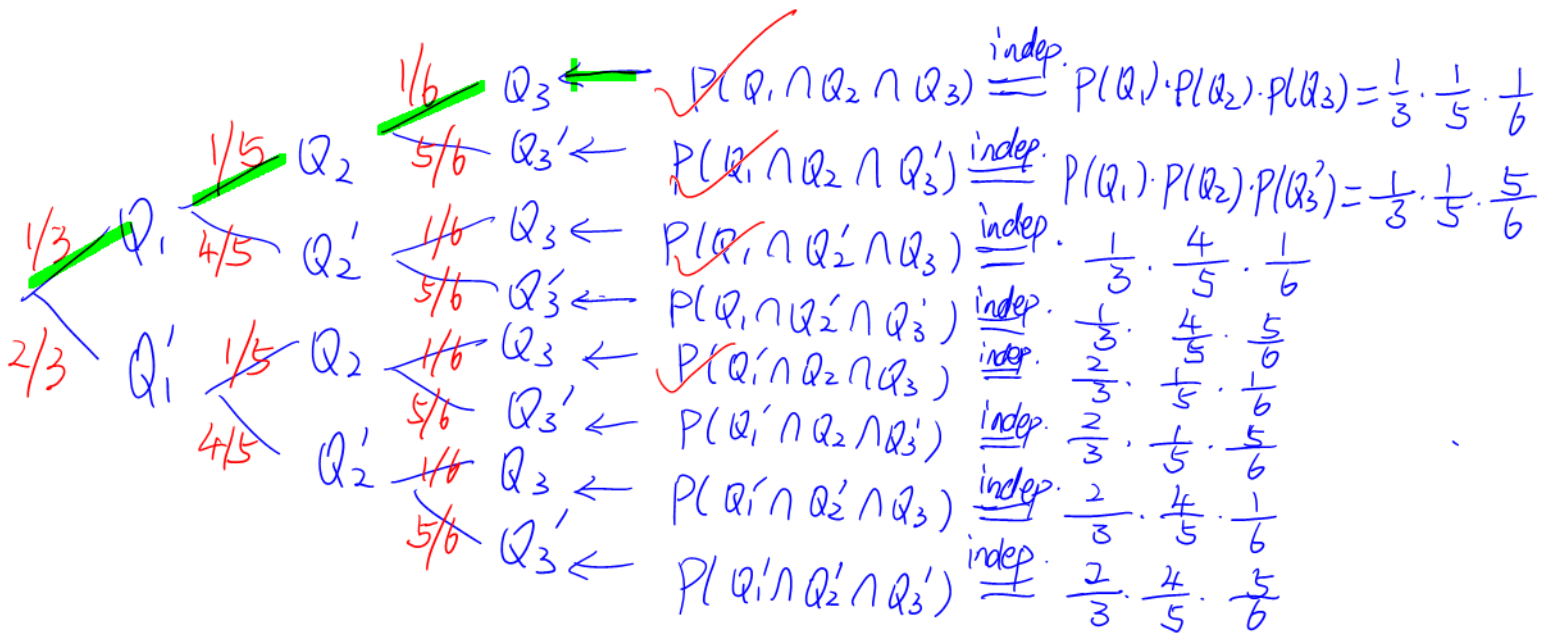


A and B independent.
 $P(A|B) = P(A)$



Example: Suppose that I have a short multiple-choice quiz to answer. Question 1 has 3 possible responses. Question 2 has 5 possible responses. Question 3 has 6 possible responses. Suppose I answer each question by randomly selecting a response. What is the probability that I will get at least two questions correct?

Q_1, Q_2 and Q_3 are independent.
 Q_1, Q_2 and Q_3 : answer is correct.



$$\begin{aligned}
 P(\text{at least 2 questions correct}) &= \frac{1}{3} \cdot \frac{1}{5} \cdot \frac{1}{6} + \frac{1}{3} \cdot \frac{1}{5} \cdot \frac{5}{6} \\
 &\quad + \frac{1}{3} \cdot \frac{4}{5} \cdot \frac{1}{6} + \frac{2}{3} \cdot \frac{1}{5} \cdot \frac{1}{6} \\
 &= \frac{12}{90}
 \end{aligned}$$

Example: Suppose that $P(A) = 0.5$, $P(B) = 0.4$, and that A and B are independent events. What is $P(A \cup B)$?

~~$$P(A \cup B) \stackrel{\text{indep.}}{=} P(A) \cdot P(B)$$~~

~~$$\neq P(A \cap B) \stackrel{\text{indep.}}{=} P(A) \cdot P(B)$$~~

$$0.5 \times 0.4$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) \stackrel{\text{indep.}}{=} P(A) + P(B) - P(A)P(B) = 0.5 + 0.4 - 0.2 = 0.7$$

Example: A boiler has five identical relief valves. The probability that any particular valve will open on demand is 0.95. Assuming independent operation of the valves, find the probability that at least one valve will open.

O_1, O_2, O_3, O_4, O_5 : valve open.
indep.

possible # of valves open: $\overset{X_1}{0}, \overset{X_2}{1}, \overset{X_3}{2}, \overset{X_4}{3}, \overset{X_5}{4}, \overset{X_6}{5}$
 $X = \text{R.V.}$

$$P(\text{at least one open}) = 1 - P(\text{"Zero" open})$$

$$= 1 - P(O_1' \cap O_2' \cap O_3' \cap O_4' \cap O_5')$$

$$\stackrel{\text{indep.}}{=} 1 - P(O_1') \cdot P(O_2') \cdots P(O_5')$$

$$= 1 - 0.05^5$$

$$\approx 0.9999997$$

$$\approx 1$$

Set 8 Random variable

For a given sample space S of some experiment, a **random variable** (rv) is any rule that associates a number with each outcome in S . In mathematical language, a random variable is a function whose domain is the sample space and whose range is the set of real numbers.

Random variables are denoted by uppercase letters, such as X and Y . We will use lowercase letters to represent some particular value of the corresponding random variable.

Example: Suppose we select a ten items at random from a production line. Let the random variable X count the number of defective items in this collection. The number of defective items could be $0, 1, \dots, 10$. Thus, X can take on the values $0, 1, \dots, 10$.

- We can now express events in terms of the values of X . $X=1$
- The probability that at least one item is defective can be expressed by $P(X \geq 1)$.
- The probability that exactly two items are defective can be expressed by $P(X = 2)$.

- Random variables can also be used for experiments where the measurements are categorical/qualitative.
- A Bernoulli random variable is a random variable which only takes on the values 0 and 1. They can be used for any experiment with only two outcomes (e.g. Yes/No, Defective/Not Defective, Pass/Fail, etc.)

Two types of Random Variables:

A discrete random variable is an rv whose possible values either constitute a finite set or else can be listed in an infinite sequence in which there is a first element, second element, and so on. ("countably" finite).

A random variable is continuous if its set of possible values consists of all numbers in a single interval on the number line.

Discrete: pmf. : $f(x) = P(X=x)$ pmf. eg. $f(0) = P(X=0)$

The probability mass function (pmf) or probability distribution of a discrete random variable $f(x)$ is defined by $f(x) = P(X = x)$ for every number x .

- ① • For finite random variables $f(x)$ can be defined by a probability distribution table.
- ② • In other cases, $f(x)$ will be defined algebraically.

Random variable

- Discrete
 - finite : # of heads tossing a coin 3 times : $\{0, 1, 2, 3\}$
 - infinite : # of customers arriving at store : $\{0, 1, 2, 3, \dots\}$
countable
- Continuous : any value in an interval.
eg: student height : $[1, 2]$

$$P(X=1) = \binom{3}{1} 0.1^1 (1-0.1)^2$$

$X \sim \text{Bin}(n=3, p=0.1)$

Example: Suppose each item on a production line has a probability of 0.1 of needing minor repairs before being sold, independently of all other items. We select three items at random.

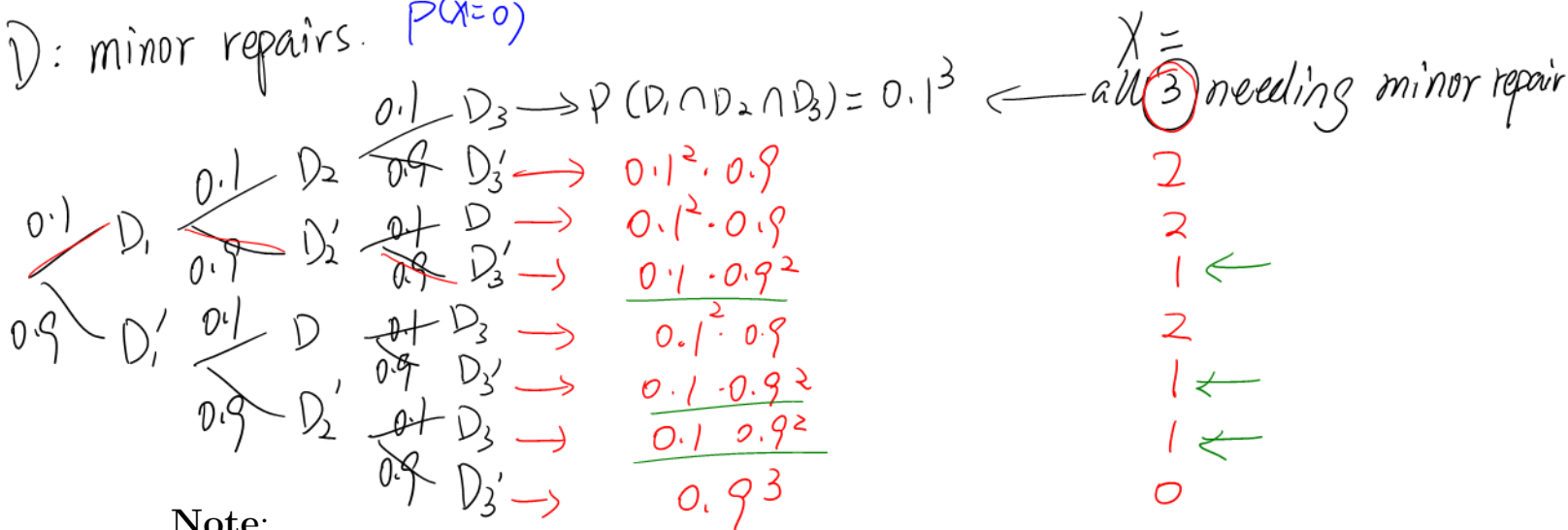
Construct the **probability distribution table** for the random variable X , where X is number of items needing minor repairs.

$X = \#$ of items needing minor repairs out of 3.

X	0	1	2	3
$f(x)$ $= P(X=x)$	0.9^3 $= 0.729$	$3 \cdot 0.1 \cdot 0.9^2$ $= 0.243$	$3 \cdot 0.1^2 \cdot 0.9$ $= 0.027$	0.1^3 $= 0.001$

$$\sum f(x) = 1$$

D : minor repairs.



Note:

- For any value of x not specified in the table, $f(x) = 0$. $\therefore f(4) = 0$
- The random variable partitions the sample space; any simple event will be associated with a value of x , and the values of x are non-overlapping.
- As a consequence, the sum of the probabilities in the table must equal 1.

$$\sum f(x) = 1$$

Example: What is the probability that no more than one item needs repairs?

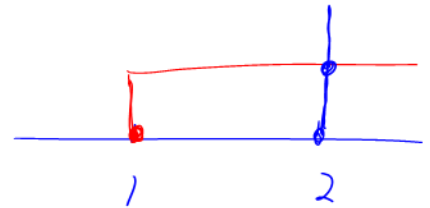
$$P(X \leq 1) = P(X=0) + P(X=1)$$

$$= 0.729 + 0.243$$

$$= 0.972$$

Example: Suppose we know that at least one item needs repairs. What is the probability that exactly two items need repairs?

$$P(X=2 | X \geq 1) = \frac{P(X=2 \cap X \geq 1)}{P(X \geq 1)}$$



$$= \frac{P(X=2)}{1 - P(X=0)}$$

$$= \frac{0.027}{1 - 0.729} = 0.0996$$

probability mass function

Sometimes, the **pmf** may depend on a **parameter**, which may be assigned any one of a number of different values. Each value of the parameter will give rise to a different **pmf**. The collection of all probability distributions for all the different values of the **parameter** is called a **family of probability distributions**.

Example: Revise our previous example, letting p be the probability that an individual item needs repairs.

Recall the probability distribution table for the random variable X , where X represents the number of items requiring minor repairs.

X	0	1	2	3
$P(X = x)$	0.9^3	$3 \cdot 0.1 \cdot 0.9^2$	$3 \cdot 0.1^2 \cdot 0.9$	0.1^3

$\leftarrow 0.1 = p$

family of probability dist^s for R.V. X .

X	0	1	2	3
$P(X=x)$	$(1-p)^3$	$3 \cdot p \cdot (1-p)^2$	$3 \cdot p^2 \cdot (1-p)$	p^3

p
↓
parameter.

pmf: $f(x) = P(X=x)$.

The cumulative distribution function (cdf), $F(x)$, of a discrete random variable X with pmf $f(x)$ is defined $F(x) = P(X \leq x)$.

$$F(5) = P(X \leq 5)$$

Example: Suppose the random variable X has the following pmf.

pmf:

X	1	2	3	4	5
f(x)	0.2	0.3	0.1	0.2	0.2

Find the cdf.

cdf

X	1	2	3	4	5
F(x)	0.2	0.5	0.6	0.8	1

↑
F(1)

pmf $\longleftrightarrow$ cdf

$$F(1) = P(X \leq 1) = P(X=1) = 0.2$$

$$F(2) = P(X \leq 2) = P(X=1) + P(X=2) = 0.5$$

$$F(3) = P(X \leq 3) = 0.6$$

$$F(4) = P(X \leq 4) = 0.8$$

Note: Although $F(x)$ gives us $P(X \leq x)$, we can also use it to find $P(X \leq x)$; $P(X = x)$, and $P(x_1 \leq X \leq x_2)$.

$P(X < x)$, $P(X \geq x)$

Example: Using our cdf to find:

- $P(X = 3) = \cancel{P(X \leq 3)} - P(X \leq 2) = 0.6 - 0.5 = 0.1$
- $P(X \geq 3) = 1 - P(X \leq 2) = 1 - 0.5 = 0.5$
- $P(2 \leq X \leq 4) = \cancel{P(X \leq 4)} - \cancel{P(X \leq 1)} = 0.8 - 0.2 = 0.6$

$$P(X < 3) = P(X \leq 2) = 0.5$$

$$P(X \geq 4) = P(X > 3) = 1 - P(X \leq 3) = 1 - 0.6 = 0.4$$

Example: Let the discrete random variable X denote the lifespan (in years) for an item produced by a particular manufacturing process. The cdf for X is the following.

cdf

X	8	9	10	11	12
$F(x)$	0.45	0.55	0.75	0.90	1.00

Example: What is the probability that the lifespan will be exactly 10 years?

$$P(X=10) = P(X \leq 10) - P(X \leq 9) = 0.75 - 0.55 = 0.2$$

Example: What is the probability the lifespan will be at least 9 years?

$$P(X \geq 9) = 1 - P(X \leq 8) = 1 - 0.45 = 0.55$$

Example: What is the probability that the lifespan will be between 9 and 11 years (inclusively)?

$$P(9 \leq X \leq 11) = P(X \leq 11) - P(X \leq 8) = 0.90 - 0.45 = 0.45$$

Example: If the item has lasted at least 10 years, what is the probability it will last no more than 11 years?

$$P(X \leq 11 \mid X \geq 10) = \frac{P(X \leq 11 \cap X \geq 10)}{P(X \geq 10)}$$

$$= \frac{P(10 \leq X \leq 11)}{1 - P(X \leq 9)}$$

$$= \frac{P(X \leq 11) - P(X \leq 9)}{1 - P(X \leq 9)} = \frac{0.9 - 0.55}{1 - 0.55} = 0.778$$



Set 9 and 10 Expected Value and Variance

Let X be a discrete random variable. The expected value, or mean, of X , denoted by μ , or by $E(X)$ is

$$\mu = E(X) = \sum x f(x)$$

x-value (pointing to x)
pmf. $f(x) = P(X=x)$ (pointing to $f(x)$)

where $f(x)$ is the pmf of X .

Interpretation: $E(X)$ is the "average" value of X . If we were to repeat the experiment many times, the long-run average of the observed values of X would approximately equal $E(X)$.

Notes:

- ① • Since $E(X)$ represents an average, it is possible (and quite common) that $E(X)$ is not one of the possible values of X . Accordingly, we do not round $E(X)$.
- ② • Note that the formula uses the probability mass function; if you only have the cdf, you will need to use it to create the pmf first before calculating $E(X)$.
cdf $\rightarrow$ pmf.
- ③ • In the case where we have several different random variables, we can use subscripts to clarify which mean we are talking about. For example, μ_X is the mean of X , while μ_Y is the mean of Y .

$$E(X) = \mu_X$$

$$E(Y) = \mu_Y$$

Example: Suppose that X has the following cumulative distribution table.

cdf

X	5	15	100
$F(x)$	$1/3$	$7/12$	1

Find $E(X)$.

$$E(X) = \sum x f(x)$$

$$= 5 \cdot \frac{1}{3} + 15 \cdot \frac{1}{4} + 100 \cdot \frac{5}{12}$$

$$= 47.08$$

pmf.

X	5	15	100
$f(x)$	$1/3$	$1/4$	$5/12$

$$f(5) = P(X=5) = P(X \leq 5) = 1/3$$

$$\begin{aligned} f(15) &= P(X=15) = P(X \leq 15) - P(X \leq 5) \\ &= \frac{7}{12} - \frac{1}{3} \\ &= \frac{1}{4} \end{aligned}$$

$$\begin{aligned} f(100) &= P(X=100) = P(X \leq 100) - P(X \leq 15) \\ &= 1 - \frac{7}{12} = \frac{5}{12} \end{aligned}$$

Repeat the experiment many times:

100, 100, 5, 15, 15, 5, 100, 15, 15, 100, - - - -

- - - - -

long-run average $\approx$ 47.08

Example: Approximately 40% of all laptops of a particular brand will need a battery replacement within 3 years of purchase. Three laptops of this brand are selected at random.

What is the expected number of laptops which will need a battery replacement within 3 years of purchase? Let X be the number of laptops needing a battery replacement within 3 years. We have the following probability distribution table:

pmf:

X	0	1	2	3
f(x)	0.216	0.432	0.288	0.064

$$E(X) = 0 \cdot 0.216 + 1 \cdot 0.432 + 2 \cdot 0.288 + 3 \cdot 0.064$$

$$= 1.2$$

0, 1, 2, 2, 1, 0, 3, - - - - - 3, 2, 0, 1, 1,

- - - - -

long-run average ≈ 1.2

Theorem: If the random variable X has a set of possible values D and pmf $f(x)$, then the expected value of any function $h(X)$, denoted by $E(h(X))$ is computed by:

$$E(h(X)) = \sum_D h(x)f(x)$$

Example: Using the pmf from the previous example, find $E(X + 2)$, and $E(X^2)$.

X	0	1	2	3
$f(x)$	0.216	0.432	0.288	0.064

$$\begin{aligned}
 E(X+2) &= E(X) + 2 \\
 &= 1.2 + 2 \\
 &= 3.2
 \end{aligned}$$

$$\begin{aligned}
 E(X+2) &= (0+2) \cdot 0.216 + (1+2) \cdot 0.432 + (2+2) \cdot 0.288 + (3+2) \cdot 0.064 \\
 &= 3.2
 \end{aligned}$$

Handwritten annotations for the second calculation: $X=0$ points to 0, $P(X=0)$ points to 0.216, $X=1$ points to 1, $P(X=1)$ points to 0.432, $X=2$ points to 2, $P(X=2)$ points to 0.288, $X=3$ points to 3, $P(X=3)$ points to 0.064.

$$\begin{aligned}
 E(X^2) &= 0^2 \cdot 0.216 + 1^2 \cdot 0.432 + 2^2 \cdot 0.288 + 3^2 \cdot 0.064 \\
 &= 2.16
 \end{aligned}$$

Note: $E(X^2)$ is not the same thing as $[E(X)]^2$.

$\uparrow$
R.V.

$\downarrow$
R.V.

$E(X) = \mu$
 $\downarrow$
constant.

Rules of Expected Value: (a, b are constants)

- $E(b) = b$; $E(5) = 5$
- $E(X + b) = E(X) + b$; $E(X + 5) = E(X) + 5$
- $E(aX) = aE(X)$; $E(5X) = 5 \cdot E(X)$

Example: If the random variable X is known to have expected value 3.8, find $E(7X + 3)$.

$$E(X) = 3.8$$

$$E(7X + 3) = E(7X) + 3 \quad \#2$$

$$= 7 \cdot E(X) + 3 \quad \#3$$

$$= 7 \cdot 3.8 + 3 = 29.6$$

Example: For the laptop experiment, the cost for a replacement battery is \$30 per laptop. What is the expected cost for our sample? (Assume that each laptop will need at most one replacement battery.)

Let $Y =$ cost per sample/customer.

$X =$ # of battery replacement.

$$Y = 30 \cdot X$$

$$E(Y) = E(30X) = 30 \cdot E(X) = 30 \cdot 1.2 = 36.$$

$$E(X) = \mu.$$

Let X have pmf $f(x)$ and expected value μ . Then the **variance** of X , denoted by $V(X)$ or σ_X^2 is:

$$\text{Variance} = E((X - \mu)^2) = \sigma^2 = V(X) = \sum_D (x - \mu)^2 f(x)$$

$h(x) = (x - \mu)^2$
The **standard deviation** of X , written $\sigma = \sqrt{\sigma^2}$.

Example: Find $V(X)$ for the laptop experiment.

x	0	1	2	3
$f(x)$	0.216	0.432	0.288	0.064

$$E(X) = 1.2 = \mu.$$

$$V(X) = \sigma_X^2 = E((X - \mu)^2) = \sum (x - \mu)^2 f(x)$$

$$= (0 - 1.2)^2 \cdot 0.216 + (1 - 1.2)^2 \cdot 0.432 + (2 - 1.2)^2 \cdot 0.288$$

$$= 0.72 + (3 - 1.2)^2 \cdot 0.064$$

$$\underline{\mu = E(X)}$$

A shortcut Formula for Variance: $V(X) = E((X - \mu)^2) = \sum (x - \mu)^2 f(x)$

$$\sigma^2 = V(X) = \sum_D x^2 f(x) - \mu^2 = E(X^2) - [E(X)]^2$$

$$\underline{V(X) = E((X - \mu)^2)} = E(X^2 - 2\mu \cdot X + \mu^2)$$

$$= E(X^2) - E(2\mu X) + E(\mu^2)$$

$$= E(X^2) - 2\mu \underline{E(X)} + \mu^2$$

$$= E(X^2) - 2\mu^2 + \mu^2$$

$$= E(X^2) - \mu^2 = \underline{\sum x^2 f(x) - \mu^2}$$

Example: Find $V(X)$ for the laptop experiment using the shortcut formula.

$$V(X) = \sigma_x^2 = E(X^2) - \mu^2 \quad \left\{ \begin{array}{l} E(X) = \mu = 1.2 \\ E(X^2) = 2.16 \end{array} \right.$$

$$= 2.16 - 1.2^2$$

$$= 0.72$$

Rules of Variance: (a, b are constants)

① • $V(b) = 0$; $V(5) = 0$

② • $V(X + b) = V(X)$; $V(X + 5) = V(X)$

③ • $V(aX) = a^2 V(X)$; $V(-5X) = (-5)^2 V(X) = 25 V(X)$

Example: If the random variable X has $E(X) = 1.9$ and $V(X) = 0.5$, find the standard deviation of $(-2X - 5)$ or $\sigma_{(-2x-5)}$.

$$V(-2X - 5) = V(-2X) + 0$$

$$= (-2)^2 V(X)$$

$$= 4 \cdot 0.5$$

$$= 2$$

$$\Rightarrow SD(-2X - 5) = \sigma_{-2X-5} = \sqrt{2}$$

Example: Find $E((-2X - 5)^2)$.

$$\left. \begin{aligned} E((-2X - 5)^2) &= V(-2X - 5) + (E(-2X - 5))^2 \\ &= 2 + (-8.8)^2 \end{aligned} \right\} \begin{aligned} V(X) &= E(X^2) - \mu^2 \\ E(X^2) &= V(X) + \mu^2 \\ E(-2X - 5) &= -2E(X) - 5 \\ &= -2 \cdot 1.9 - 5 \\ &= -8.8 \end{aligned}$$

Set 11 The Binomial Probability Distribution

A binomial experiment is a probability experiment that satisfies the following four requirements:

- ① • There must be a fixed number of trials, n . *① fixed n .*
- ② • Each trial can have only two possible outcomes. These outcomes can be considered as either success (S) or failure (F). *② two possible outcomes.*
- ③ • The trials are independent, so that the outcomes on any particular trial does not influence the outcome on any other trial. *③ independent trials.*
- ④ • The probability of a success $P(S) = p$ must remain the same for each trial. *④ $p(S)=p$ for all trials.*

If a random variable X counts the number of successes out of n trials in a binomial experiment, then X is a binomial random variable, and we write $X \sim \text{Bin}(n; p)$, where n is the number of trials, and p is the probability of successes.

$$X \sim \text{Bin}(n, p)$$

of successes out of n . *# of trials* *$p(S)=p$*

$$\{0, 1, 2, \dots, n\}$$

Example: In a manufacturing process, each item has a probability of 0.05 of being defective, independent of all other items. Suppose 12 items are selected at random, and we let W denote the number of defective items.

Binomial experiment {

- ①. fixed n , $n=12$ ✓
- ②. two possible outcomes ✓
- ③. independent trials ✓
- ④. $p(s) = 0.05$ ✓

$$W \sim \text{Bin}(n=12, p=0.05)$$

Example: On a multiple choice test, there are 10 questions, each with 8 possible responses. I will complete the test by randomly selecting answers. Let X count the number of questions I get correct.

Binomial experiment {

- ①. fixed n , $n=10$ ✓
- ②. two possible outcomes ✓
- ③. independent ✓
- ④. $p(s) = 1/8$ ✓

$$X \sim \text{Bin}(n=10, p=1/8)$$

Example: Each randomly selected person has a 40% probability of having type A blood. We select and test people until we find 3 people with type A blood. Let Y count the number of people we must select.

Not Binomial experiment {

- ①. fixed n ✗
- ②. two possible outcomes ✓
- ③. independent ✓
- ④. $p(s) = 0.4$ ✓

Example: We have a box containing 5 red and 5 green marbles. We select 3 marbles from the box, one at a time, without replacement. Let B count the number of red marbles selected.

Not Binomial experiment {

- ①. fixed n , $n=3$ ✓
- ②. two possible outcomes ✓
- ③. independent ✗
- ④. $p(s) = p$ ✗

Binomial Probability Distribution function:

$$X \sim \text{Bin}(n, p)$$

$$P(X = x) = \binom{n}{x} p^x (1-p)^{n-x} \quad \text{for } x = 0, 1, 2, \dots, n$$

prob. of successes
 prob. of failures.
 "n choose x"
 "nC x."

Example: In the manufacture of lithium batteries, it is found that 7% of all batteries are defective. Suppose that we test 6 randomly selected batteries. What is the probability that at least two batteries are defective?

Let $X = \#$ of defective batteries out of 6.

$$[0, 1, 2, \dots, 6] \leftarrow X \sim \text{Bin}(n=6, p=0.07)$$

$$P(X \geq 2) = 1 - P(X=0) - P(X=1)$$

$$= 1 - \binom{6}{0} 0.07^0 (1-0.07)^{6-0} - \binom{6}{1} 0.07^1 (1-0.07)^{6-1}$$

$$= 0.0608$$

Expected Value and Variance: If $X \sim B(n, p)$, then:

$$E(X) = np \text{ and } V(X) = np(1 - p)$$

Example: What is the expected number of defective lithium batteries per batch of 6?

$$X \sim \text{Bin}(n=6, p=0.07)$$

$$E(X) = np = 6 \cdot 0.07 = 0.42$$

Example: For the battery-testing experiment, what is the variance?

$$V(X) = np(1-p) = 6 \cdot 0.07(1-0.07) = 0.3906.$$

Example: If we have $X \sim B(n=20, p=0.2)$, then $P(X \leq 10) = ?$

$$P(X \leq 10) = P(X=0) + P(X=1) + \dots + P(X=10)$$

Cumulative Distribution Tables: These tables give $P(X \leq x)$ for "nice" values of n and p .

$$X \sim \text{Bin}(20, 0.2)$$

$$P(X \leq 10) = 0.9994.$$

Example: It is known that 30% of all laptops of a certain brand experience hard-drive failure within 3 years of purchase. Suppose that 20 laptops are selected at random. Let the random variable X denote the number of laptops which have experienced hard-drive failure within 3 years of purchase.

$$X \sim \text{Bin}(n=20, p=0.3)$$

- (a) What is the probability that no more than 6 laptops will experience hard-drive failure?

$$P(X \leq 6) = 0.6080$$

- (b) What is the probability that exactly 6 or 7 laptops will experience hard-drive failure?

$$\begin{aligned} P(6 \leq X \leq 7) &= P(X=6) + P(X=7) \\ &= P(X \leq 7) - P(X \leq 5) = 0.7723 - 0.4164 = 0.3559 \end{aligned}$$

- (c) What is the probability that at least 4 laptops will experience hard-drive failure?

$$P(X \geq 4) = 1 - P(X \leq 3) = 1 - 0.1071 = 0.8929$$

- (d) Given that at least 3 laptops experience hard-drive failure, what is the probability that no more than 6 laptops will experience hard-drive failure?

$$\begin{aligned} P(X \leq 6 | X \geq 3) &= \frac{P(X \leq 6 \cap X \geq 3)}{P(X \geq 3)} \\ &= \frac{P(3 \leq X \leq 6)}{1 - P(X \leq 2)} = \frac{P(X \leq 6) - P(X \leq 2)}{1 - P(X \leq 2)} \end{aligned}$$



CUTOFF: Test 1.

$$= \frac{0.6080 - 0.0355}{1 - 0.0355} = 0.5936$$

Set 12 The Poisson Probability Distribution

A **Poisson probability distribution** is useful for describing the number of events that will occur during a specific interval of time or in a specific distance, area, or volume.

Examples:

- ① • The number of traffic accidents at a particular intersection.
- ② • The number of house fire claims per month that are received by an insurance company.
- ③ • The number of people who are infected with the AIDS virus in a certain neighbourhood.
- ④ • The number of people who walk into a barber shop without an appointment.

Poisson Random Variable: If in a Poisson experiment, X counts the number of successes that occur in a period of time/space, then X is a Poisson random variable. We write $X \sim \text{Poisson}(\lambda)$, where λ is the average number of successes per region/interval.

↓
"lambda"

Poisson Probability Distribution: If $X \sim \text{Poisson}(\lambda)$, then

$$P(X = x) = \frac{e^{-\lambda} \lambda^x}{x!} \quad \text{for } x = 0, 1, 2, 3, \dots$$

$$0! = 1$$

Example: Suppose a machine makes defective items according to a Poisson process at an average rate of 5 defective items per hour. What is the probability that the machine will make exactly 4 defective items in an hour?

Let X denote the number of defective items made by the machine in an hour.

$$X \sim \text{Poisson}(\lambda = 5/\text{hr})$$

$$P(X = 4) = \frac{e^{-5} \cdot 5^4}{4!} = 0.1755$$

Expected Value and Variance: If $X \sim \text{Poisson}(\lambda)$, then:

$$E(X) = \lambda \text{ and } V(X) = \lambda$$

Example: What is the expected number of defective items made by the machine in an hour? What is the variance?

$$E(X) = V(X) = \lambda = 5$$

Cumulative Distribution Tables: These tables give $P(X \leq x)$ for "nice" values of λ .

Example: Suppose the machine is watched for three hours. What is the probability that it will make no more than 12 defective items?

Recall that the machine makes on average 5 defective items per hour. This means that it will make on average 15 defective items per three hour period.

Let X denote the number of defective items made in a three hour period.

$$\underline{X} \sim \text{Poisson}(\lambda = 3 \cdot 5 = 15 / 3\text{-hr period})$$

$$P(X \leq 12) = 0.2676$$

Example: What is the probability that at least 6 defective items will be made during the three hour period?

$$\begin{aligned} P(X \geq 6) &= 1 - P(X \leq 5) \\ &= 1 - 0.0028 \\ &= 0.9972 \end{aligned}$$

Example: What is the probability that exactly 13 defective items will be made during the three hour period?

$$\begin{aligned} P(X = 13) &= \frac{e^{-15} 15^{13}}{13!} \\ &= \cancel{0.1121} \dots \cancel{12} \dots 13 \\ &= P(X \leq 13) - P(X \leq 12) \\ &= 0.3632 - 0.2676 = 0.0956 \end{aligned}$$

Example: Suppose it is known that at a particular intersection, there are an average of 2 car accidents per week. If there is at least one accident this week, what is the probability that there will be at least three accidents in total this week? Assume that the occurrence of accidents follows a Poisson process.

Let X denote the number accidents in a week.

$$X \sim \text{Poisson}(\lambda = 2)$$

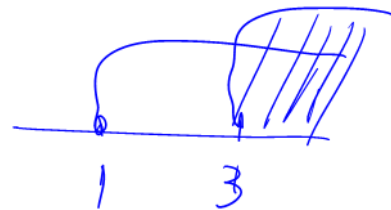
$$P(X \geq 3 | X \geq 1) = \frac{P(X \geq 3 \cap X \geq 1)}{P(X \geq 1)}$$

$$= \frac{P(X \geq 3)}{1 - P(X \leq 0)}$$

$$= \frac{1 - P(X \leq 2)}{1 - P(X \leq 0)}$$

$$= \frac{1 - 0.6767}{1 - 0.1353}$$

$$= 0.3739$$



assume # of errors follows
 $\downarrow$ a Poisson distⁿ.

Example: Suppose that a typist makes on average of 2 errors per page.
 Suppose the typist is creating a ten-page document.

What is the probability that exactly three of the pages "do not contain any errors"?

Let $Y = \#$ of pages "do not contain any errors".

$S =$ no errors 1, 2, 3, 4, ... 10
 $F =$ some errors. $\begin{matrix} \diagup & \diagdown \\ S & F \end{matrix}$ $\begin{matrix} \diagup & \diagdown \\ S & F \end{matrix}$ $\begin{matrix} \diagup & \diagdown \\ S & F \end{matrix}$ $\begin{matrix} \diagup & \diagdown \\ S & F \end{matrix}$... $\begin{matrix} \diagup & \diagdown \\ S & F \end{matrix}$

$$Y \sim \text{Bin}(n=10, p = P(S) = P(\text{no errors in one page}))$$

$$P(Y=3) = \binom{10}{3} p^3 (1-p)^7$$

$$= \binom{10}{3} 0.1353^3 (1-0.1353)^7$$

$$= 0.1074$$

Let $X = \#$ of errors in one page.

$$X \sim \text{Poisson}(\lambda=2)$$

$$P(X=0) = p \quad \text{at } \lambda=2$$

$$p = 0.1353$$

Poisson approximation to Binomial: If X is a binomial random variable where n is very large and p is very small then X can be approximated with a Poisson distribution with $\lambda = np$.

$$X \sim \text{Bin}(n, p) \xrightarrow{n \geq 100, np \leq 10} X \overset{\text{approximated.}}{\sim} \text{Poisson}(\lambda = np)$$

Provided $n \geq 100$ and $np \leq 10$, the approximation will be quite good. It will still be reasonably good when $n \geq 20$, as long as $p \leq 0.05$.

Example: Brugada syndrome is a rare disease which afflicts 0.0002% of the population. Suppose 10,000 people are selected at random and tested for Brugada syndrome. What is the probability that no more than 3 of the tested people will have Brugada syndrome?

Let $X = \#$ of tested people will have the disease out of 10,000 people.

$$X \sim \text{Bin}(n=10,000, p=0.0002)$$

$$\begin{aligned} P(X \leq 3) &= P(X=0) + P(X=1) + \dots + P(X=3) \\ &= \binom{10000}{0} 0.0002^0 (1-0.0002)^{10000} + \dots \end{aligned}$$

$$n=10,000 \geq 100 \quad \checkmark \quad np=10000 \cdot 0.0002 = 2 \leq 10 \quad \checkmark$$

$$X \sim \text{Poisson}(\lambda = np = 2)$$

$$\begin{aligned} P(X \leq 3) \quad \text{at } \lambda = 2 \\ = 0.8571 \end{aligned}$$

Set 13 and 14 Continuous Probability Distributions

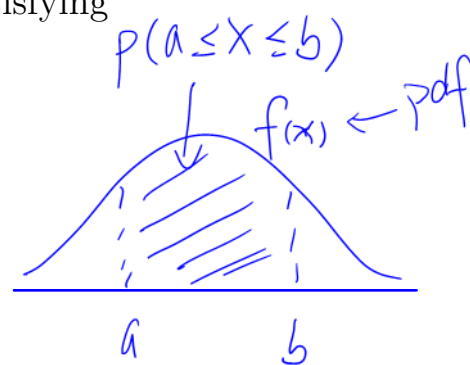
Continuous Random Variable: A random variable is one that can assume an uncountable number of values.

For a continuous random variable, the probability distribution or probability density function (pdf) is a function $f(x)$ satisfying

$$P(a \leq X \leq b) = \int_a^b f(x) dx$$

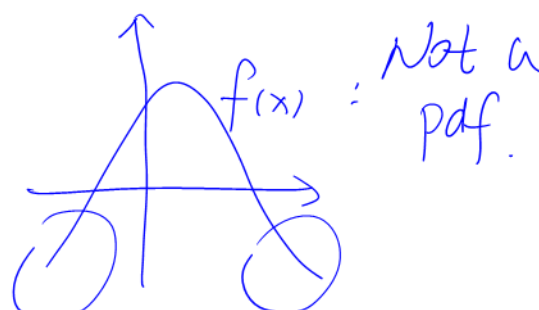
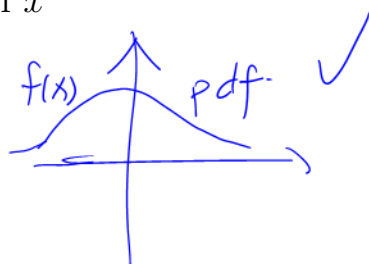
pdf

for any two numbers a and b (with $a \leq b$).



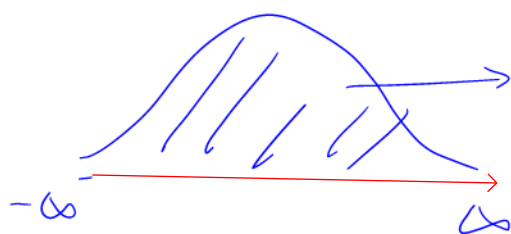
Requirements for a Probability Density Function: The following requirements apply to a probability density function $f(x)$ whose range is $a \leq X \leq b$.

- ① • $f(x) \geq 0$ for all x



- $\int_{-\infty}^{\infty} f(x) dx = \text{area under the entire graph of } f(x) = 1$

②.



$$P(-\infty < X < \infty) = \int_{-\infty}^{\infty} f(x) dx = 1$$

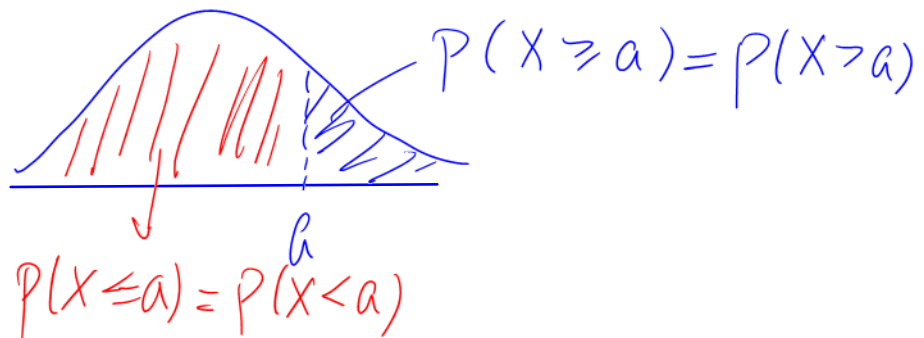
①
Note: Since a valid pdf must never be below the x axis, we can interpret $P(a \leq X \leq b)$ as the area under $f(x)$ on the interval $[a, b]$.

Only for continuous R.V. Some consequences for any valid pdf:

1. $P(X = a) = 0$ for any a .

R.V. $P(X = a) = \lim_{\Delta x \rightarrow 0} P(a - \Delta x < X < a + \Delta x) = \lim_{\Delta x \rightarrow 0} \int_{a - \Delta x}^{a + \Delta x} f(x) dx = 0$

2. $P(X \geq a) = P(X > a)$



3. $P(X \leq a) = P(X < a)$

4. $P(X \geq a) = 1 - P(X \leq a)$

or $P(X \geq a) = 1 - P(X < a)$

$P(X > a) = 1 - P(X < a)$

$P(X > a) = 1 - P(X \leq a)$

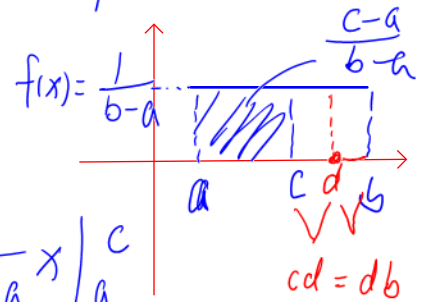
Uniform Probability Distribution:

A continuous random variable X is said to have a uniform distribution on the interval $[a, b]$ if the pdf of X is

$$X \sim \text{unif}[a, b]$$

pdf:

$$f(x) = \begin{cases} \frac{1}{b-a} & a \leq X \leq b \\ 0 & \text{otherwise} \end{cases}$$



$$\begin{aligned} P(a \leq X \leq c) &= \int_a^c f(x) dx = \int_a^c \frac{1}{b-a} dx = \frac{1}{b-a} x \Big|_a^c \\ &= \frac{c-a}{b-a} \end{aligned}$$

$$P(a \leq X \leq c) = (c-a) \cdot \frac{1}{b-a}$$

Example: Suppose the weekly output of a steel mill is a uniformly distributed random variable that lies between 110 and 175 metric tons. What is the probability that the steel mill will produce more than 150 metric tons next week?

Let X = weekly output of this steel mill
 $X \sim \text{Unif}[110, 175]$

$$\Rightarrow f(x) = \frac{1}{175-110} = \frac{1}{65}$$

$$P(X > 150) = \int_{150}^{175} f(x) dx = \int_{150}^{175} \frac{1}{65} dx$$

$$= \frac{1}{65} x \Big|_{150}^{175}$$

$$= \frac{1}{65} \cdot (175-150) = \frac{25}{65}$$

Example: Find the value k so that $f(x)$ is a valid pdf.

$$f(x) = \begin{cases} kx^3 & 2 \leq X \leq 5 \\ 0 & \text{otherwise} \end{cases}$$

①. $f(x) = kx^3 \geq 0 \quad 2 \leq X \leq 5 \Rightarrow k \geq 0$

②. $\int_{-\infty}^{\infty} f(x) dx = 1 \Rightarrow \int_2^5 kx^3 dx = 1$

$$k \cdot \frac{1}{4} x^4 \Big|_2^5 = 1$$

$$k \cdot \frac{1}{4} (5^4 - 2^4) = 1$$

$$f(x) = \frac{4}{609} x^3 \quad 2 \leq X \leq 5$$

$$k = \frac{4}{609} \cdot \geq 0 \quad \checkmark$$

Example: With the above pdf, find $P(3 \leq X \leq 4)$.

$$P(3 \leq X \leq 4) = \int_3^4 f(x) dx = \int_3^4 \frac{4}{609} \cdot x^3 dx$$

$$= \frac{4}{609} \cdot \frac{1}{4} x^4 \Big|_3^4$$

$$= \frac{1}{609} (4^4 - 3^4)$$

$$= \frac{175}{609}$$

Example: Find an expression for $P(X \leq b)$, where b is some number in $[2, 5]$.

$$\begin{aligned}
 \underline{P(X \leq b)} &= \int_{-\infty}^b f(x) dx = \int_2^b \frac{4}{609} x^3 dx \\
 \underline{F(b)} &= \\
 \text{cdf.} &= \frac{4}{609} \cdot \frac{1}{4} x^4 \Big|_2^b \\
 &= \frac{1}{609} (b^4 - 16) \\
 &= \frac{1}{609} b^4 - \frac{16}{609}.
 \end{aligned}$$

Note: We can conclude that:

- ① • $P(X \leq x) = x^4/609 - 16/609$ for all x in the interval $[2, 5]$.
- ② • If x is less than 2, $P(X \leq x) = 0$. i.e. $P(X \leq 1) = 0$
- ③ • If x is greater than 5, $P(X \leq x) = 1$. i.e. $P(X \leq 6) = 1$

Using this, we have the cumulative distribution function (cdf), $F(x)$, for the given pdf.

$$\underline{\underline{F(x) = \begin{cases} 0 & X < 2 \\ x^4/609 - 16/609 & 2 \leq X \leq 5 \\ 1 & X > 5 \end{cases}}}$$

pdf

$$\text{cdf: } F(x) = \int_{-\infty}^x f(t) dt \quad \text{i.e. } F(x) = \int_2^x \frac{4}{609} t^3 dt.$$

Note: The fundamental theorem of calculus tells us that for every x at which $F'(x)$ exists, that $F'(x) = f(x)$.

(cdf)' pdf.

Example: Suppose the random variable X has the following cdf:

$$F(x) = \begin{cases} 0 & X < 0 \\ x/(x+1) & X \geq 0 \end{cases}$$

Find the pdf for the random variable X .

$$\begin{aligned} f(x) = F'(x) &= \frac{\frac{d}{dx}[x] \cdot (x+1) - \frac{d}{dx}[x+1] \cdot x}{(x+1)^2} \\ &= \frac{1 \cdot (x+1) - 1 \cdot x}{(x+1)^2} \\ &= \frac{1}{(x+1)^2} \end{aligned}$$

pdf:
$$f(x) = \begin{cases} \frac{1}{(x+1)^2} & x \geq 0 \\ 0 & \text{o/w.} \end{cases}$$

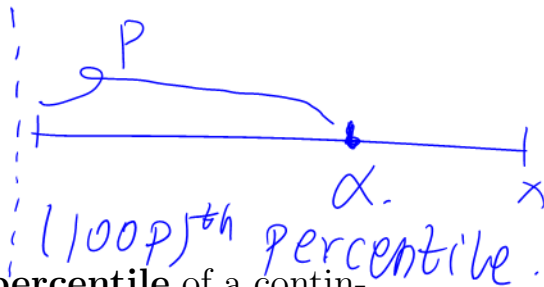
ex: median = 50th percentile.

$$0.5 = P(X \leq \alpha) : \alpha = \text{median}$$

Percentiles of a Continuous Distribution:

Let p be a number between 0 and 1. The $(100p)^{\text{th}}$ percentile of a continuous random variable is the value α such that $F(\alpha) = p$.

$$= P(X \leq \alpha)$$



Example: For the random variable from the previous example, find the 90th percentile.

$$\text{cdf: } F(x) = \frac{x}{x+1} \quad x \geq 0$$

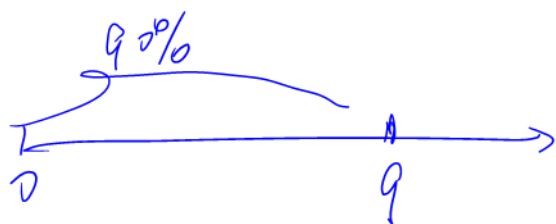
$$90^{\text{th}} \text{ percentile} \Rightarrow F(\alpha) = 0.9 = P(X \leq \alpha)$$

$$\frac{\alpha}{\alpha+1} = 0.9$$

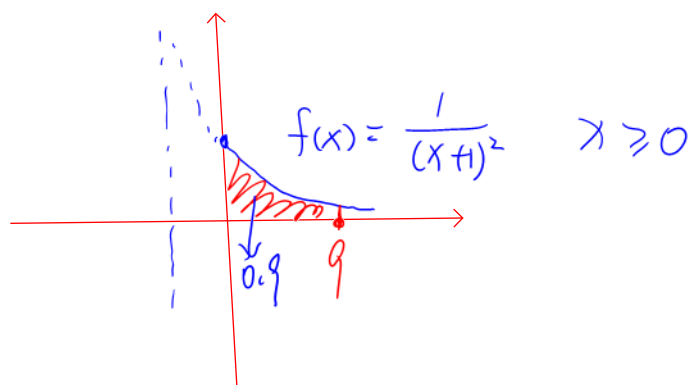
$$\alpha = 0.9\alpha + 0.9$$

$$0.1\alpha = 0.9$$

$$\alpha = 9$$

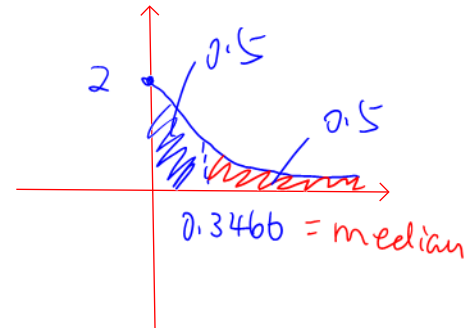


$\nwarrow$ 90th percentile.



Example: Suppose the random variable X has pdf:

pdf:
$$f(x) = \begin{cases} 2e^{-2x} & X \geq 0 \\ 0 & \text{otherwise} \end{cases}$$



Find the median of the distribution.

Note: The median of a continuous random variable is the 50th percentile.

Find cdf first:

$$\begin{aligned} F(x) &= \int_{-\infty}^x f(t) dt = \int_0^x 2 \cdot e^{-2t} dt \\ &= 2 \cdot \left(-\frac{1}{2}\right) e^{-2t} \Big|_0^x \\ &= -1 \cdot (e^{-2x} - e^{-2 \cdot 0}) \\ &= 1 - e^{-2x} \end{aligned}$$

$$F(x) = 0.5 = 1 - e^{-2x}$$

$$\hookrightarrow \text{median} \Rightarrow e^{-2x} = 0.5$$

$$-2x = \ln 0.5$$

$$x = \ln 0.5 / -2$$

$$x = 0.3466$$

Expected Value and Variance:

The **expected value** or **mean** of a continuous random variable X with pdf $f(x)$ is:

$$E(X) = \mu = \int_{-\infty}^{\infty} x f(x) dx$$

(Handwritten: pdf. with an arrow pointing to f(x) and a wavy line under x)

(provided this interval converges)

The **variance** of a continuous random variable X with pdf $f(x)$ is:

$$V(X) = \sigma^2 = \int_{-\infty}^{\infty} (x - \mu)^2 f(x) dx$$

(provided this interval converges)

and the **standard deviation**, $\sigma = \sqrt{\sigma^2}$.

As before, we have the shortcut formula:

$$V(X) = E(X^2) - [E(X)]^2 = E(X^2) - \mu^2$$

Note: The proof of the computation formula for the continuous case is almost identical to that for the discrete case simply use integrals instead of summations.

$$V(X) = E(X^2) - \mu^2$$

$$= \int_{-\infty}^{\infty} x^2 f(x) dx - \mu^2$$

Integration by parts:

Example: Suppose the random variable X has pdf:

$$f(x) = \begin{cases} 2e^{-2x} & X \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

$$\int u dv = u \cdot v - \int v du$$

Find the mean and the variance.

$$E(X) = \int_{-\infty}^{\infty} x \cdot f(x) dx = \int_0^{\infty} x \cdot 2 \cdot e^{-2x} dx$$

$$= 2 \left[x \cdot \left(-\frac{1}{2} e^{-2x}\right) \Big|_0^{\infty} - \int_0^{\infty} -\frac{1}{2} e^{-2x} dx \right] \begin{cases} u = x \Rightarrow du = dx \\ dv = e^{-2x} dx \\ v = -\frac{1}{2} e^{-2x} \end{cases}$$

$$= 2 \left[0 + \frac{1}{2} \cdot \left(-\frac{1}{2} e^{-2x} \Big|_0^{\infty}\right) \right]$$

$$= 2 \left[\frac{1}{2} \cdot \left(-\frac{1}{2}\right) (e^{-2\infty} - 1) \right] = \frac{1}{2} = \mu$$

$$V(X) = E(X^2) - \mu^2$$

$$E(X^2) = \int_0^{\infty} x^2 \cdot 2e^{-2x} dx$$

$$\begin{cases} u = x^2 \Rightarrow du = 2x dx \\ dv = e^{-2x} dx \Rightarrow \\ v = -\frac{1}{2} e^{-2x} \end{cases}$$

$$= 2 \left[x^2 \cdot \left(-\frac{1}{2} e^{-2x}\right) \Big|_0^{\infty} - \int_0^{\infty} -\frac{1}{2} e^{-2x} \cdot 2x dx \right]$$

$$= 2 \left[0 + \frac{1}{2} \int_0^{\infty} \underbrace{x \cdot 2e^{-2x}}_{u = \frac{1}{2}} dx \right]$$

$$= 2 \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{2}$$

$$\Rightarrow V(X) = \frac{1}{2} - \left(\frac{1}{2}\right)^2 = \frac{1}{4}$$

Set 15 and 16 The Normal Distributions

The **normal distribution** is the most important one in all of probability and statistics. Many numerical populations have distributions that can be fit very closely by an appropriate normal curve.

Normal Density Function: If X is normally distributed with mean μ and standard deviation σ , then we write $X \sim N(\mu, \sigma)$. The pdf of X is:

pdf

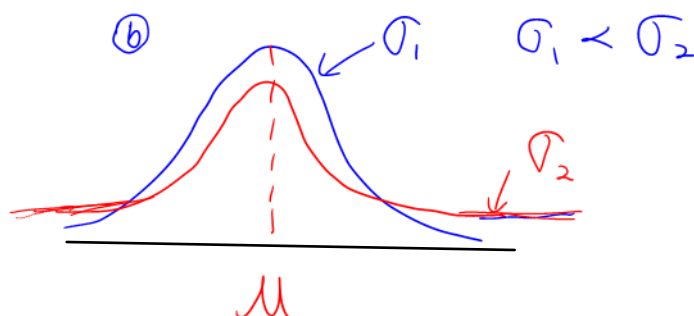
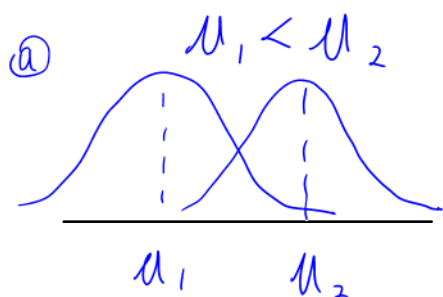
$$f(x; \mu, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/(2\sigma^2)} \quad -\infty < X < \infty$$

parameters.

Properties of Normal Curves:



- ① • All normal curves are defined on $(-\infty, \infty)$ and is bell-shaped.
- ② • There is a single peak at $x = \mu$ and the curve is symmetric about this peak.
- ③ • The mean, median, and mode are all μ .
- ⓐ • As μ increases, the peak moves further to the right. As μ decreases, the peak moves further to the left. (μ is a location parameter)
- ⓑ • As σ increases, the peak becomes lower, and the curve becomes flatter. As σ decreases, the curve becomes more peaked, and the peak becomes taller. (σ is a **scale parameter**).

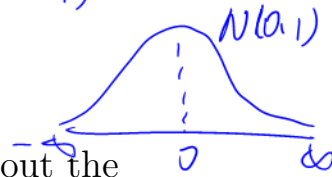


$N(0, 2)$, $N(15, 100)$, $N(5, 1)$ - - - -

Standard Normal Distribution: The standard normal random variable has mean 0 and standard deviation 1. We use the letter Z to denote the standard normal distribution.

$$Z \sim N(\mu=0, \sigma=1)$$

$$f(z; 0, 1) = \frac{1}{\sqrt{2\pi}} e^{-(z)^2/(2)} \quad -\infty < Z < \infty$$



The standard normal curve has its peak at 0, and is symmetric about the y-axis.

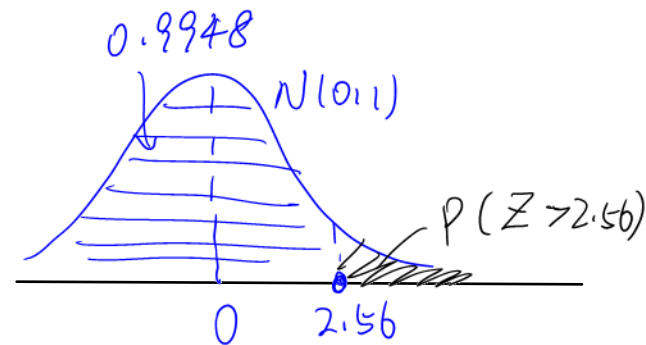
Cumulative Standardized Normal Probabilities

If our random variable is Z , then we denote the cdf $P(Z \leq z)$ by $F(Z)$.

Example: Find $P(Z \leq 2.56)$ $= \int_{-\infty}^{2.56} f(z) dz$ $\times$

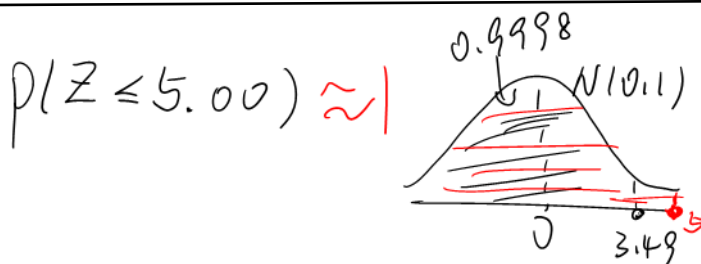
$P(Z \leq 2.56) = 0.9948$

Annotations: A red arrow points to the inequality sign. Below the number 2.56, a red bracket labeled "row" spans the digits 5 and 6. Below the digit 2, a red arrow labeled "column" points to the circled number 1. Below the digit 5, a red arrow points to the circled number 2.



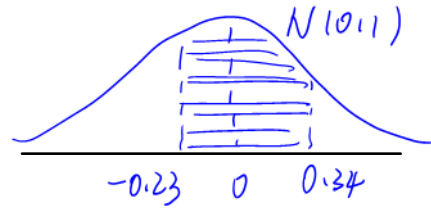
$$P(Z > 2.56) = 1 - P(Z \leq 2.56)$$

$$= 1 - 0.9948 = 0.0052$$



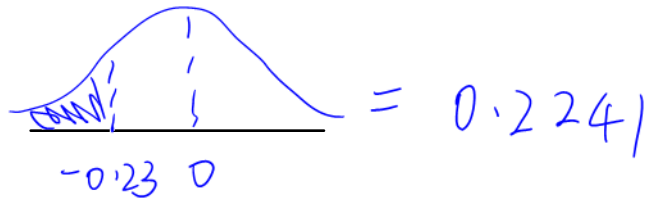
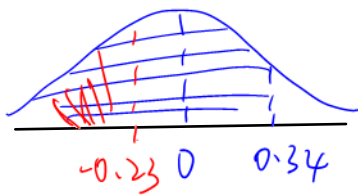
$$P(Z \leq -5.00) \approx 0$$

Example: Find $P(-0.23 \leq Z \leq 0.34)$



$$P(-0.23 \leq Z \leq 0.34)$$

$$= P(Z \leq 0.34) - P(Z \leq -0.23) = 0.6331 - 0.4090$$

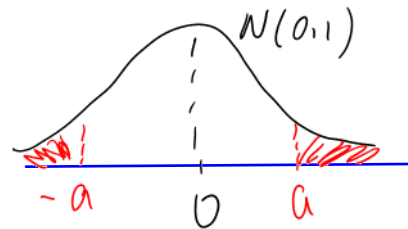


Symmetry Property: Since the random variable Z is symmetric about $Z = 0$, then for any α :

$$P(Z \leq -\alpha) = P(Z \geq \alpha)$$

or

$$P(Z \geq -a) = P(Z \leq a)$$

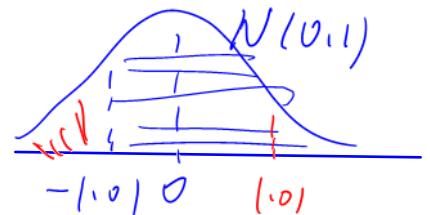


Example: Find $P(Z \geq -1.01)$

$$= P(Z \leq 1.01) = 0.8438$$

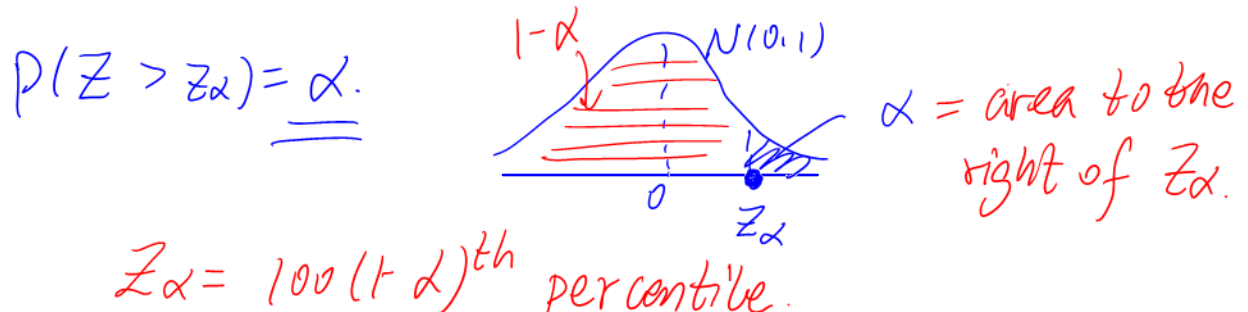
$$\text{or, } P(Z \geq -1.01) = 1 - P(Z \leq -1.01)$$

$$= 1 - 0.1562 = 0.8438$$

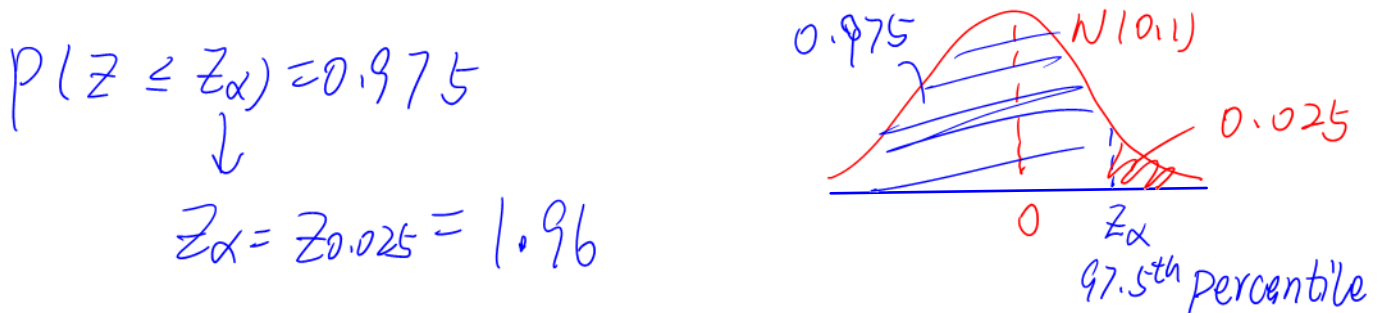


$$P(Z \leq K) = \underline{0.8438} \quad K = ? \underline{1.01}$$

Notation: z_α is the number such that $P(Z > z_\alpha) = \alpha$. Alternately, z_α is the $100(1 - \alpha)$ percentile of the standard normal distribution.



Example: Find the 97.5th percentile of the standard normal distribution.



Standardizing a Normal Random Variable: If X is normally distributed with mean μ and standard deviation σ , then:

$$X \sim N(\mu, \sigma)$$

$$Z = \frac{x - \mu}{\sigma}$$

$$\downarrow$$

$$Z = \frac{X - \mu}{\sigma} \sim N(0, 1)$$

For $X \sim N(\mu, \sigma)$, we can standardize as follows:

$$P(a \leq X \leq b) = P\left(\frac{a - \mu}{\sigma} \leq \frac{X - \mu}{\sigma} \leq \frac{b - \mu}{\sigma}\right) = P\left(\frac{a - \mu}{\sigma} \leq Z \leq \frac{b - \mu}{\sigma}\right)$$

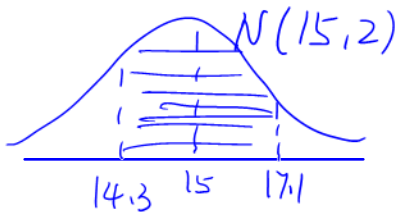
$$P(a \leq X \leq b) = P\left(\frac{a - \mu}{\sigma} \leq \frac{X - \mu}{\sigma} \leq \frac{b - \mu}{\sigma}\right) = P\left(\frac{a - \mu}{\sigma} \leq Z \leq \frac{b - \mu}{\sigma}\right)$$

Example: The masses of a certain type of bolt is approximately normally distributed with $\mu = 15$ g, and $\sigma = 2$ g. What is the probability that a randomly selected bolt has a mass between 14.3 g and 17.1 g?

Let X denote the mass of a randomly selected bolt.

Then $X \sim N(\mu = 15; \sigma = 2)$.

$$P(14.3 \leq X \leq 17.1) = P\left(\frac{14.3 - 15}{2} \leq \frac{X - \mu}{\sigma} \leq \frac{17.1 - 15}{2}\right)$$

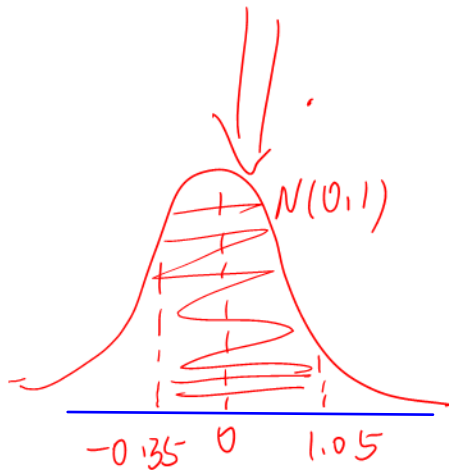


$$= P(-0.35 \leq Z \leq 1.05)$$

$$= P(Z \leq 1.05) - P(Z \leq -0.35)$$

$$= 0.8531 - 0.3632$$

$$= \underline{\underline{0.4899}}$$



Example: What is the probability that at a randomly selected bolt will have a mass of at least 20 g?

$$X \sim N(\mu=15, \sigma=2)$$

$$P(X \geq 20) = P\left(\frac{X-\mu}{\sigma} \geq \frac{20-15}{2}\right)$$

$$= P(Z \geq 2.5)$$

$$= 1 - P(Z < 2.5)$$

$$= 1 - 0.9938 = 0.0062$$

Example: What is the minimum mass of the heaviest 5% of bolts?

$$X \sim N(15, 2)$$

$$P(X \geq k) = 0.05$$

$$\Rightarrow P\left(\frac{X-\mu}{\sigma} \geq \frac{k-15}{2}\right) = 0.05$$

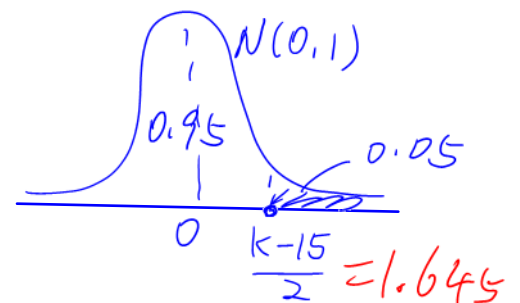
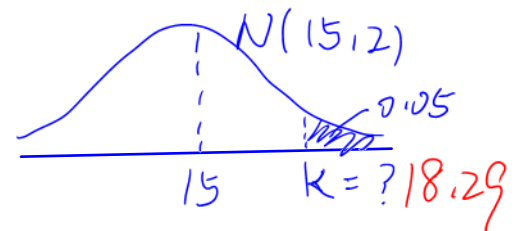
$$\Rightarrow P\left(Z \geq \frac{k-15}{2}\right) = 0.05$$

$$\Rightarrow P\left(Z \leq \frac{k-15}{2}\right) = 0.95$$

$$P(Z \leq 1.645) = 0.95$$

$$\frac{k-15}{2} = 1.645$$

$$k = 18.29$$



$$\left\{ \begin{array}{l} P(Z \leq 1.64) = 0.9495 \\ P(Z \leq 1.65) = 0.9505 \\ P(Z \leq 1.645) = 0.95 \end{array} \right.$$

Approximating the Binomial Distribution with the Normal Distribution: Suppose $X \sim \text{Bin}(n, p)$ where np and $n(1-p)$ are both at least 5 (i.e., the expected number of successes and the expected number of failures are both at least 5). Then X can be approximated with a normal distribution, with a mean of np , and a standard deviation of $\sqrt{np(1-p)}$.

$$X \sim \text{Bin}(n, p) \xrightarrow{np \geq 5 \text{ and } n(1-p) \geq 5} X \overset{\text{Approx}}{\sim} N(\mu=np, \sigma=\sqrt{np(1-p)})$$

Then we have:

$$P(X \leq x) \approx P\left(Z \leq \frac{x - np}{\sqrt{np(1-p)}}\right)$$

$$Z = \frac{X - np}{\sqrt{np(1-p)}} \sim N(0, 1)$$

Example: Suppose it is known that 20% of batteries have a lifespan shorter than the advertised lifespan. Suppose that 100 batteries are selected at random. What is the approximate probability that at least 10 batteries will have a short lifespan?

Let X denote the number of batteries with a short lifespan. Then $X \sim \text{Bin}(100, 0.2)$. We note that both np and $n(1-p)$ exceed 5.

$$X \sim \text{Bin}(n=100, p=0.2) \text{ and } np=20, n(1-p)=80 \geq 5. \checkmark$$

$$X \sim N(\mu=np=20, \sigma=\sqrt{np(1-p)}=\sqrt{100 \cdot 0.2(1-0.2)}=4)$$

$$P(X \geq 10) = P\left(\frac{X - np}{\sqrt{np(1-p)}} \geq \frac{10 - 20}{4}\right)$$

$$= P\left(\overset{\parallel}{Z} \geq -2.5\right) = 1 - P(Z \leq -2.5)$$

$$= 1 - 0.0062 = 0.9938$$

Example: Suppose it is known that the reaction time of type of voice-activated robot is normally distributed with $\mu = 6.3$ microseconds, and $\sigma = 2$ microseconds.

Let $X =$ the reaction time of a robot.

Suppose I select one voice-activated robot at random. What is the probability that its reaction time is between 5.6 and 7 microseconds?

$$X \sim N(\mu = 6.3, \sigma = 2)$$

$$\begin{aligned} P(5.6 < X < 7) &= P\left(\frac{5.6 - 6.3}{2} < \frac{X - \mu}{\sigma} < \frac{7 - 6.3}{2}\right) \\ &= P(-0.35 < Z < 0.35) \\ &= P(Z < 0.35) - P(Z < -0.35) \\ &= 0.6368 - 0.3632 = 0.2736 \end{aligned}$$

Example: Suppose that I select five robots and test each of them. Assume the reaction time of each robot is independent of the other robot reaction-times. What is the probability that exactly three of the robots will have a reaction time between 5.6 and 7 microseconds?

Let $Y =$ # of "successful" robot out of 5.

1. 2. 3. 4. 5. $\hookrightarrow$ reaction time b/w 5.6 and 7.

$\begin{matrix} \diagup & \diagdown & \diagup & \diagdown & \diagup \\ S & F & S & F & S \end{matrix}$ $P(S) = P(5.6 < X < 7) = 0.2736$

$$Y \sim \text{Bin}(n=5, P=P(S)=0.2736)$$

$$\begin{aligned} P(Y=3) &= \binom{5}{3} 0.2736^3 (1 - 0.2736)^2 \\ &= 0.1081 \end{aligned}$$

Set 17 The Exponential Distribution

The density curve corresponding to any normal distribution is bell-shaped and therefore symmetric. There are many practical situation in which the variable of interest to an investigator might have a skewed distribution. One family of distributions that has this property is the **Exponential distribution**.

The Exponential distribution: The random variable X has exponential distribution with parameter λ ($\lambda > 0$) if the pdf is:

$$X \sim \text{Exp}(\lambda) \quad \text{pdf: } f(x) = \begin{cases} \lambda e^{-\lambda x} & X \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

We can find $E(X)$ and $V(X)$ by integrating:

$$E(X) = 1/\lambda \text{ and } V(X) = 1/\lambda^2.$$

$$\mu = E(X) = \int_0^{\infty} x \cdot f(x) dx = \int_0^{\infty} x \cdot \lambda e^{-\lambda x} dx = \frac{1}{\lambda}$$

$$V(X) = E(X^2) - \mu^2 = \int_0^{\infty} x^2 \lambda e^{-\lambda x} dx - \mu^2 = \frac{1}{\lambda^2}$$

The **pdf** of the exponential distribution can be easily integrated, giving us the **cdf**:

$$P(X \leq x) = F(x) = \begin{cases} 1 - e^{-\lambda x} & X \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

$$F(x) = \int_0^x f(t) dt = \int_0^x \lambda e^{-\lambda t} dt = 1 - e^{-\lambda t}$$

Note: the exponential distribution is usually used to model waiting times.

Example: During the lunch hour, the average waiting time to use an automatic bank machine is 6 minutes. It is assumed that the random variable X which counts the number of minutes a customer waits before service has exponential distribution.

What is the probability that a customer will need to wait at least 9 minutes?

let $x = \text{waiting time}$.

$$X \sim \text{Exp}(\lambda) \text{ and } E(X) = 6 = \frac{1}{\lambda} \Rightarrow \lambda = \frac{1}{6}$$

$$\text{pdf: } f(x) = \frac{1}{6} e^{-x/6} \quad x \geq 0$$

$$\text{cdf: } F(x) = 1 - e^{-x/6} \quad x \geq 0$$

$$\begin{aligned} P(X \geq 9) &= \int_9^{\infty} f(x) dx = \int_9^{\infty} \frac{1}{6} e^{-x/6} dx = -e^{-x/6} \Big|_9^{\infty} \\ &= e^{-9/6} = 0.2231 \end{aligned}$$

or:

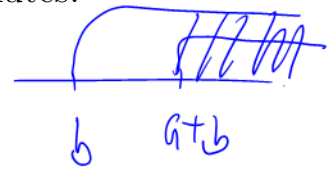
$$\begin{aligned} P(X \geq 9) &= 1 - P(X < 9) = 1 - F(9) \\ &= 1 - (1 - e^{-9/6}) \\ &= e^{-9/6} = 0.2231 \end{aligned}$$

The **memoryless property**: Suppose $X \sim \text{Exp}(\lambda)$. Then for any $a, b \geq 0$:

$$P(X \geq a + b | X \geq b) = P(X \geq a)$$

This means that the probability of a person needing to wait at least a minutes more if they've already waited b minutes, is **the same** as the probability of a newly-arrived person needing to wait a minutes.

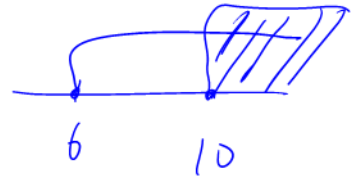
$$\underline{P(X \geq a + b | X \geq b) = \frac{P(X \geq a + b \cap X \geq b)}{P(X \geq b)}}$$



$$\begin{aligned} \text{cdf: } \bar{F}(x) &= 1 - e^{-\lambda x} \\ &= \frac{P(X \geq a + b)}{P(X \geq b)} \\ &= \frac{1 - P(X < a + b)}{1 - P(X < b)} \\ &= \frac{1 - (1 - e^{-\lambda(a+b)})}{1 - (1 - e^{-\lambda b})} \\ &= \frac{e^{-\lambda(a+b)}}{e^{-\lambda b}} = e^{-\lambda a} \cdot \cancel{e^{-\lambda b}} \\ &= e^{-\lambda a} \\ &= 1 - (1 - e^{-\lambda a}) \\ &= 1 - \bar{F}(a) \\ &= 1 - P(X \leq a) \\ &= \underline{P(X \geq a)} \end{aligned}$$

Example: Suppose you've already been waiting to use the bank machine for six minutes. What is the probability your total waiting time will be at least 10 minutes?

$$X \sim \text{Exp}(\lambda = \frac{1}{6})$$



$$P(X \geq 10 | X \geq 6) = \frac{P(X \geq 10 \cap X \geq 6)}{P(X \geq 6)}$$

$$= \frac{P(X \geq 10)}{P(X \geq 6)}$$

$$F(x) = \frac{1}{6} e^{-\frac{x}{6}} \quad x \geq 0$$

$$= \frac{1 - P(X < 10)}{1 - P(X < 6)}$$

$$= \frac{1 - (1 - e^{-10/6})}{1 - (1 - e^{-6/6})} = e^{-4/6} \approx 0.2231$$

$$P(X \geq 10 | X \geq 6) = P(X \geq 4 + 6 | X \geq 6)$$

$$\stackrel{\text{memoryless}}{=} P(X \geq 4)$$

$$= 1 - P(X < 4)$$

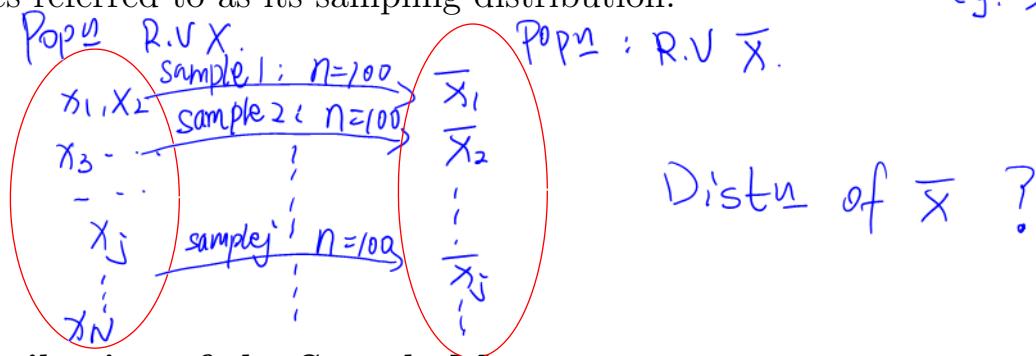
$$= 1 - (1 - e^{-4/6})$$

$$= e^{-4/6} \approx 0.2231$$

X is a R.V. | $\bar{X} = \frac{X_1 + X_2 + \dots + X_n}{n}$
 $\hookrightarrow$ is a R.V.!

Set 18 The Central Limit Theorem

Sampling Distribution: The probability distribution of a statistic is sometimes referred to as its sampling distribution. eg. $\bar{x}$.



The distribution of the Sample Mean

0. The random variables $X_1, X_2, X_3, \dots, X_n$ are **independent** and **identically distributed (iid)**, another way of saying this X 's are independent of one another and arise from the same probability distribution.

$\Rightarrow$ **1.** The random variables $X_1, X_2, X_3, \dots, X_n$ are iid from a **population** with mean μ and standard deviation σ . Then $\Rightarrow E(X) = \mu, V(X) = \sigma^2$

- The mean of the sample mean, $E(\bar{X}) = \mu_{\bar{X}} = \mu$

$$E(\bar{X}) = E\left(\frac{X_1 + X_2 + \dots + X_n}{n}\right) = \frac{1}{n} [E(X_1) + E(X_2) + \dots + E(X_n)]$$

$\xleftarrow{\mu} \quad \quad \quad \xleftarrow{\mu} \quad \quad \quad \xleftarrow{\mu}$

$$= \frac{1}{n} n \cdot \mu = \mu.$$

- The variance of the sample mean, $V(\bar{X}) = \sigma_{\bar{X}}^2 = \sigma^2/n$ and the standard deviation of the sample mean: $\sigma_{\bar{X}} = \sigma/\sqrt{n}$

$$V(\bar{X}) = V\left(\frac{X_1 + X_2 + \dots + X_n}{n}\right) = \frac{1}{n^2} [V(X_1) + V(X_2) + \dots + V(X_n)]$$

$\xrightarrow{\text{independent}}$

$$= \frac{1}{n^2} (\sigma^2 + \sigma^2 + \dots + \sigma^2)$$

$$= \frac{1}{n^2} \cdot n\sigma^2 = \frac{\sigma^2}{n}$$

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$$

→ **2. Proposition:** a linear combinations of normal random variables are normal: Consider $X_1, X_2, X_3, \dots, X_n$ are independent random variables from a normal distribution with mean μ and standard deviation σ , then for any n , $\bar{X} \sim N(\mu, \sigma/\sqrt{n})$.

① $X \sim N(\mu, \sigma)$ $\xrightarrow{\text{any "n"}}$ $\bar{X} \sim N(\mu, \sigma/\sqrt{n})$

"CLT"

3. The Central Limit Theorem: Random variables $X_1, X_2, X_3, \dots, X_n$ are iid from a population with mean μ and standard deviation σ , then if $n \geq 30$, $\bar{X} \stackrel{\text{Approx}}{\sim} N(\mu, \sigma/\sqrt{n})$

Note: The standard deviation of $\bar{X}$, $\sigma_{\bar{X}}$, is called the standard error of the mean.

② if $n \geq 30$ $\xrightarrow{\text{any dist}^n \text{ of } X}$ $\bar{X} \sim N(\mu, \sigma/\sqrt{n})$

Example: A random sample of size $n = 40$, is from a Poisson distribution with $\lambda = 100$. Find the probability that the sample mean, $\bar{X}$, at least 103.

$X \sim \text{Poisson}(\lambda = 100)$

$E(X) = \lambda = 100 = \mu$

$V(X) = \lambda = 100 = \sigma^2$

$n = 40$, by CLT.

$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$

$P(\bar{X} \geq 103) = ?$

$\bar{X} \sim N(\mu, \sigma/\sqrt{n})$

$\downarrow \quad \downarrow$
 $100 \quad \sqrt{100}/\sqrt{40}$

$= P\left(\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \geq \frac{103 - 100}{\sqrt{100}/\sqrt{40}}\right)$

$= P(Z \geq 1.90) = 1 - P(Z < 1.90)$

$= 1 - 0.9713 = 0.0287$

Example: Suppose at a factory, it is known that the levels of liquid in soda bottles is normally distributed, with a mean of 355 mL, and standard deviation of 2 mL. Let x_1, x_2, x_3, x_4 denote liquid content of four randomly selected bottles. Find the probability that the average liquid content will be less than 356 mL.

$$n=4.$$

$$X \sim N(\mu=355, \sigma=2)$$

$$P(\bar{X} < 356) = ?$$

↓
Distⁿ of $\bar{X}$?

$$\bar{X} \sim N(\mu=355, \sigma/\sqrt{n}=2/\sqrt{4})$$

$$P(\bar{X} < 356) = P\left(\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} < \frac{356 - 355}{2/\sqrt{4}}\right)$$

$$= P(Z < 1)$$

$$= 0.8413$$

Another Application of the Central Limit Theorem:

Suppose we have nominal/categorical data and are interested in observations in a particular category.

Notation: For population proportion (the parameter) we use p . For sample proportion (the statistic) we use $\hat{p}$

$$\hat{p} \rightarrow p$$

$$\bar{x} \rightarrow \mu$$

For a sample of size n , if there are X observations in the particular category, then $\hat{p} = X/n$.

Example: We are interested in p , the proportion of all customers who enter our store and make a purchase. We randomly select 175 customers who enter our store; of these, 64 make a purchase.

We calculate that $\hat{p}$ the proportion in our sample who make a purchase, is

$$\hat{p} = 64/175.$$

Distⁿ of $\hat{p}$?

Recall: A binomial variable, $X \sim \text{Bin}(n, p)$ where np and $n(1-p)$ are both at least 5, the $X \sim N(np, \sqrt{np(1-p)})$.

$X \sim \text{Bin}(n, p)$ $np \geq 5$ $n(1-p) \geq 5$ $X \sim N(np, \sqrt{np(1-p)})$ $\downarrow E(X)$ $\downarrow \sigma_x$
 $\Rightarrow$ The estimator of p is $\hat{p} = X/n$. Since $\hat{p}$ is just X multiplied the constant $1/n$, therefore $\hat{p} \sim N(\mu_{\hat{p}} = p, \sigma_{\hat{p}} = \sqrt{p(1-p)/n})$

$$\frac{X}{n} = \hat{p} \sim N(\mu_{\hat{p}} = p, \sigma_{\hat{p}} = \sqrt{p(1-p)/n})$$

$$E(\hat{p}) = E\left(\frac{X}{n}\right) = \frac{1}{n} E(X) = \frac{1}{n} \cdot np = p$$

$$V(\hat{p}) = V\left(\frac{X}{n}\right) = \frac{1}{n^2} V(X) = \frac{1}{n^2} \cdot np(1-p) = p(1-p)/n$$

$$Z = \frac{\hat{p} - p}{\sqrt{\frac{p(1-p)}{n}}} \sim N(0, 1)$$

Example: It is known that 18% of all savings accounts at a bank are overdrawn at least once per year. Suppose that we select 500 accounts at random. What is the probability that the sample proportion of these 500 accounts will be greater than 0.17?

$$\text{b/c : } np = 500 \cdot 0.18 = 90 \geq 5$$

$$n(1-p) = 500(1-0.18) = 410 \geq 5$$

$$\Rightarrow \hat{p} \sim N(p, \sqrt{p(1-p)/n})$$

$$P(\hat{p} > 0.17) ?$$

$\downarrow$

Distrib of $\hat{p}$?

$$: p = 0.18, n = 500$$

$$P(\hat{p} > 0.17) = P\left(\frac{\hat{p} - p}{\sqrt{p(1-p)/n}} > \frac{0.17 - 0.18}{\sqrt{0.18(1-0.18)/500}}\right)$$

$$= P(Z > -0.58)$$

$$= 1 - P(Z < -0.58)$$

$$= 0.7190$$

Set 19 Confidence Interval

- ① **Point Estimate:** An estimate of the value of an unknown parameter (i.e. the mean, the variance), using a single value.

$$\hat{p} \rightarrow P : \bar{x} \rightarrow \mu$$

$$s^2 \rightarrow \sigma^2$$

- ② **Interval Estimate:** An estimate of the value of an unknown parameter using an interval of plausible values.

2a.) **Confidence Interval:** A confidence interval is calculated by first selecting a confidence level, which is a measure of the degree of reliability of the interval.

For example: A 95% confidence interval for the parameter μ is an random interval (L, U) constructed by a procedure that satisfies

$$P(L < \mu < U) = 0.95.$$

The 0.95 is called the **confidence level**, and L and U are the lower and upper confidence limits.

i.e. 95% confidence interval.

$$95\% = 100(1-\alpha)\%$$

$$\alpha = 0.05$$

In general, the $100(1-\alpha)\%$ confidence interval for μ (assuming σ is known)

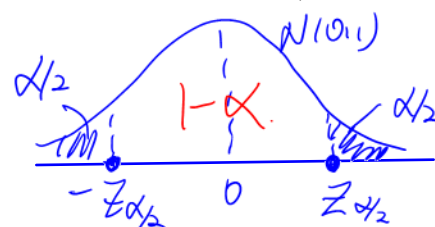
$$P(-Z_{\alpha/2} < Z < Z_{\alpha/2}) = 1-\alpha$$

by CLT: $Z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$

$$\Rightarrow P(-Z_{\alpha/2} < \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} < Z_{\alpha/2}) = 1-\alpha$$

$$\Rightarrow P(-Z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}} < \bar{x} - \mu < Z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}) = 1-\alpha$$

$$\Rightarrow P(-\bar{x} - Z_{\alpha/2} \frac{\sigma}{\sqrt{n}} < -\mu < -\bar{x} + Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}) = 1-\alpha$$



$$\Rightarrow P(\bar{x} - Z_{\alpha/2} \frac{\sigma}{\sqrt{n}} < \mu < \bar{x} + Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}) = 1-\alpha$$

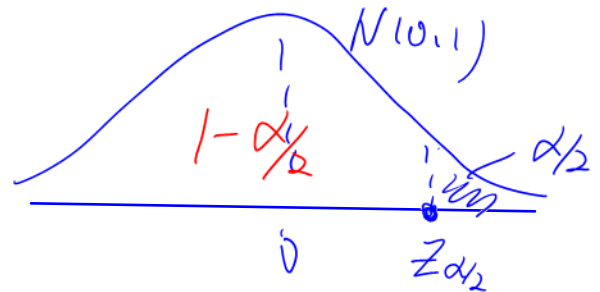
$$P(L < \mu < U) = 1-\alpha$$

therefor, $100(1-\alpha)\%$ CI for μ : (by CLT)

$$\bar{x} \pm z_{\alpha/2} * \frac{\sigma}{\sqrt{n}}$$

where $z_{\alpha/2}$ is the critical value for an $100(1-\alpha)\%$ confidence interval.

$$P(Z > z_{\alpha/2}) = \alpha/2$$



$$\Rightarrow P(Z \leq z_{\alpha/2}) = 1 - \alpha/2$$

$\hookrightarrow$ Z-score for the cumulative prob = $1 - \alpha/2$

Example: What is the critical value for an 80% confidence interval?

$$100(1-\alpha)\% = 80\% \Rightarrow \alpha = 0.2$$

$$\alpha/2 = 0.1$$

$$z_{\alpha/2} = z_{0.1} = 1.28$$



Other common critical values:

% Confidence	Critical Value
$\alpha/2=0.05 \Rightarrow 90$	$z_{\alpha/2}=z_{0.05}=1.645$
$\alpha=0.05 \Rightarrow 95$	$z_{\alpha/2}=z_{0.025}=1.96$
$\alpha=0.01 \Rightarrow 99$	$z_{\alpha/2}=z_{0.005}=2.575$

$$\textcircled{1} X \sim N(\mu, \sigma) \Rightarrow \bar{x} \sim N(\mu, \sigma/\sqrt{n})$$

$$\textcircled{2} n > 30$$

Example: Suppose that for 25 people, the average height $\bar{x} = 1.7$ m. Suppose it is known that the standard deviation of people's heights is 0.08 m. Find a 95% confidence interval for μ , the population's mean height.

$n=25$

Assume People's height follows a normal distn.

$$\sigma = 0.08$$

$$\begin{aligned} \alpha &= 0.05 \\ \alpha/2 &= 0.025 \\ Z_{\alpha/2} &= Z_{0.025} \\ &= 1.96 \\ \uparrow \\ \text{Z-score} \\ \text{for cumulative} \\ \text{prob.} &= 1 - \alpha/2 \\ &= 1 - 0.025 \\ &= 0.975 \end{aligned}$$

$$100(1-\alpha)\%$$

$$\leftarrow 95\% \text{ CI for } \mu: \bar{x} \pm Z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}$$



$$Z_{\alpha/2} = Z_{0.025} = 1.96$$

$$= 1.7 \pm 1.96 \cdot \frac{0.08}{\sqrt{25}}$$

$$= 1.7 \pm 0.03136$$

$$= (1.66864, 1.73136)$$

We are 95% confident that the popu mean, μ is between 1.66864 and 1.73136.

Repeat the experiment 1000 times: (Suppose $\mu = 1.7$)
check CI contains " μ "

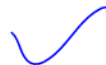
$$S_1: [1.67, 1.73]$$

$$S_2: [1.65, 1.71]$$

$$\vdots$$

$$S_{10}: [1.62, 1.69]$$

$$\vdots$$



Approx.

95% CI's

contain " μ "

Interpretation: If we were to repeat this experiment many times, 95% of the confidence intervals we create would contain the true value of μ .

Recall the $100(1 - \alpha)\%$ confidence interval for μ (assuming σ is known)

$$\bar{x} \pm \boxed{z_{\alpha/2} * \frac{\sigma}{\sqrt{n}}} \leftarrow \text{margin of error.}$$

where $z_{\alpha/2} * \frac{\sigma}{\sqrt{n}}$ is called the margin of error (MOE).

$$\text{margin of error} \rightarrow d = z_{\alpha/2} * \frac{\sigma}{\sqrt{n}}$$

Determining sample size: Suppose we wish to estimate μ within d units of the sample mean.

$$n = \left(\frac{z_{\alpha/2} * \sigma}{d} \right)^2$$

$$\text{eg: } n = 21.1 \rightarrow n = 22$$

$$n = 21.9 \rightarrow n = 22.$$

n : always round up to the next integer.

Example: The lifespan of a brand of tablet computers is normally distributed with a standard deviation of 3 months. How many tablets should we sample to estimate μ within 1 month of the mean with 95% confidence?

$$\sigma = 3$$

$$95\% \Rightarrow Z_{\alpha/2} = Z_{0.025} = 1.96 \quad \sigma = 3$$

$$\text{within} = \text{margin of error} = d = 1$$

$$n = \left(\frac{Z_{\alpha/2} \cdot \sigma}{d} \right)^2 = \left(\frac{1.96 \cdot 3}{1} \right)^2 = 34.57$$

$$n = 35$$

Example: How many tablets should we sample to estimate μ within 0.5 months of the mean with 95% confidence?

$$n = \left(\frac{Z_{\alpha/2} \cdot \sigma}{d} \right)^2 = \left(\frac{1.96 \cdot 3}{0.5} \right)^2$$

$$= 138.3$$

$$n = 139$$

Set 20 and 21 Hypothesis Test I about a population mean

Example: A certain medication is supposed to contain 350 mg of the active ingredient per pill. It is known from previous work that this content is normally distributed with a standard deviation of $\sigma = 3.5 \text{ mg}$. Suppose a random sample of 5 pills are taken, and the average content is 346.4 mg.

$$X \sim N(\mu, \sigma)$$

$$n = 5$$

$$= \mu$$

$$\bar{X} = 346.4$$

Is there evidence that the mean medication content is greater than 350 mg?

claim: $\mu = 350$

want to verify: $\mu > 350$

Recall the methods to estimate a parameter: **Point Estimate, Interval Estimate.**

Another method is the **Hypothesis Test.**

Define the Hypothesis test:

null hypothesis and alternative hypothesis:

H_0 : claim (or opposite H_1) H_1 : (H_a): (want to verify)

$H_0: \mu = 350$ (or $\mu \leq 350$)

(assume H_0 is true)

$H_1: \mu > 350$

Note: The null hypothesis, H_0 is the commonly accepted fact; it is the opposite of the alternate hypothesis H_1 which researchers want to verify.

The p-value: is the probability of seeing results as extreme as the observations actually made if the null hypothesis were true.

The p-value: a measure of how rare is the observed result if H_0 is true.

The smaller the p-value the stronger evidence against H_0

Report the strength of evidence against H_0 :

- Very strong if $p\text{-value} \leq 0.01$
 - Strong if $0.01 < p\text{-value} \leq 0.05$
 - Weak if $0.05 < p\text{-value} \leq 0.10$
 - Little or none if $0.10 < p\text{-value}$
- (Little to none)

Errors and Hypothesis Tests: Two types of errors are possible in a hypothesis test.

Decision	H_0 True	H_0 False
Retain H_0	✓	Type II error
Reject H_0	Type I error	✓

Big mistake. ↓
 Type I error = $P(\text{Reject } H_0 \mid H_0 \text{ True})$
 Type II error = $P(\text{Retain } H_0 \mid H_0 \text{ False})$

Example continuous: (Step by Step)

1. Define the parameter of interest.

Let μ = mean medication content per pill.

2. Properly specify the null and alternative hypotheses. (Hypotheses in the form of sentences are unacceptable.)

$$H_0: \mu \leq 350 \quad H_1: \mu > 350$$

↓ (or $\mu = 350$) (want to verify)

assume H_0 is true.

3. What test statistic are you using? What probability distribution does the test statistic have?

Test Statistic: is a function of the sample data used as a basis of deciding whether H_0 should be rejected.

① $X \sim N(\mu, \sigma)$
② $n > 30 \Rightarrow$ By CLT: $\bar{X} \sim N(\mu, \sigma/\sqrt{n})$

test-stats: $Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \sim N(0, 1)$

↖ if H_0 is true

4. Calculate the observed value of the test statistic.

$\bar{X} = 346.4$
 $n = 5$
 $\sigma = 3.5$

↘ the distr. of the test statistic if H_0 is true.

$$Z_{obs} = \frac{346.4 - 350}{3.5/\sqrt{5}} = -2.299 \approx -2.30$$

5. Calculate the p-value.

Since $H_1: \mu > 350$

$$p\text{-value} = P(Z > Z_{obs}) = P(Z > -2.30)$$

$$= 1 - P(Z < -2.30) = 1 - 0.0107 = 0.9893$$

6. What level of evidence do we have against the null hypothesis? (Very Strong / Strong / Weak / Little or None)?

Since $p\text{-value} = 0.9893 > 0.1$, we have little or none evidence against H_0 .

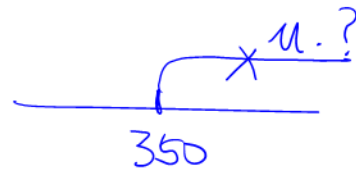
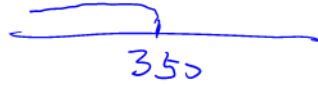
Another approach to draw conclusion: the significant level, α ✓ Given

- ① • If $p\text{-value} \leq \alpha$, then **reject** the null hypothesis.
- ② • If $p\text{-value} > \alpha$, then **don't reject** the null hypothesis

Example continuous: if the significant level $\alpha=0.05$, what would our conclusion be?

Since $p\text{-value} = 0.9893 > \alpha = 0.05$,
We don't reject H_0 .

or/ eg: $H_1: \mu < 350$ one-tailed test.



Note: The test we just performed is called a one-tailed test, as $H_1:$

$\mu > 350$

The two-tailed test: If the question ask: Is there evidence that the mean

medication content is ~~greater than~~ different from 350 mg? *want to verify $\mu \neq 350$.*

1. let $\mu = \dots$

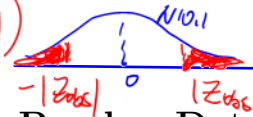
2. $H_0: \mu = 350$ $H_1: \mu \neq 350$

3. test stats: $Z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$
by CLT: $\sim N(0,1)$

$$4. Z_{obs} = \frac{346.4 - 350}{3.5/\sqrt{5}} = -2.30$$

5. since $H_1: \mu \neq 350 \Leftarrow$ two tailed.

$$P\text{-value} = P(Z \leq -|Z_{obs}| \text{ or } Z \geq |Z_{obs}|) \\ = 2 \cdot P(Z < -|Z_{obs}|)$$



Three forms of Alternative Hypothesis and P-value Determination

$\downarrow$ H_0 is true.

$H_0: \mu \leq \mu_0$ $H_1: \mu > \mu_0 \longrightarrow P\text{-value} = P(Z > Z_{obs})$

$H_0: \mu \geq \mu_0$ $H_1: \mu < \mu_0 \longrightarrow P\text{-value} = P(Z < Z_{obs})$

$H_0: \mu = \mu_0$ $H_1: \mu \neq \mu_0 \longrightarrow P\text{-value} = 2 \cdot P(Z < -|Z_{obs}|)$

$H_0: \mu = \mu_0$

Set 22 Hypothesis Test II about a population mean

Recall that if the samples come from a normal distribution, or if we have sample size $n \geq 30$, and σ is known, then:

$$\frac{\bar{x} - \mu}{\sigma/\sqrt{n}} = Z \sim N(0, 1)$$

What if σ is unknown?

Case I: We have a very large ($n \geq 30$) observations. In this case, s will be a good estimate for σ .

$$\frac{\bar{x} - \mu}{s/\sqrt{n}} = Z \sim N(0, 1)$$

The estimated standard error (e.s.e) of $\bar{X}$: $s/\sqrt{n}$

Tests and Intervals for μ : If we have $n \geq 30$, with σ unknown:

Test statistic:

$$Z = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

Confidence intervals:

$$\bar{x} \pm z_{\alpha/2} * \frac{s}{\sqrt{n}}$$

Case II: Small sample size ($n < 30$), but the samples come from a normal (or approximately normal) distribution.

$$\frac{\bar{x} - \mu}{s/\sqrt{n}} = t_{n-1}$$

t_{n-1} is the t **distribution** with $n - 1$ is the **degrees of freedom**.

Properties of the t distributions:

- It is continuous and defined for all real numbers.
- It is symmetric, bell-shaped, and centered at zero.
- The number of degrees of freedom affect the shape of the distribution; as the number of degrees of freedom increases, the distribution becomes more peaked and the tails become thinner.
- When the number of degrees of freedom is large, the t-distribution is close to being a standard normal distribution.

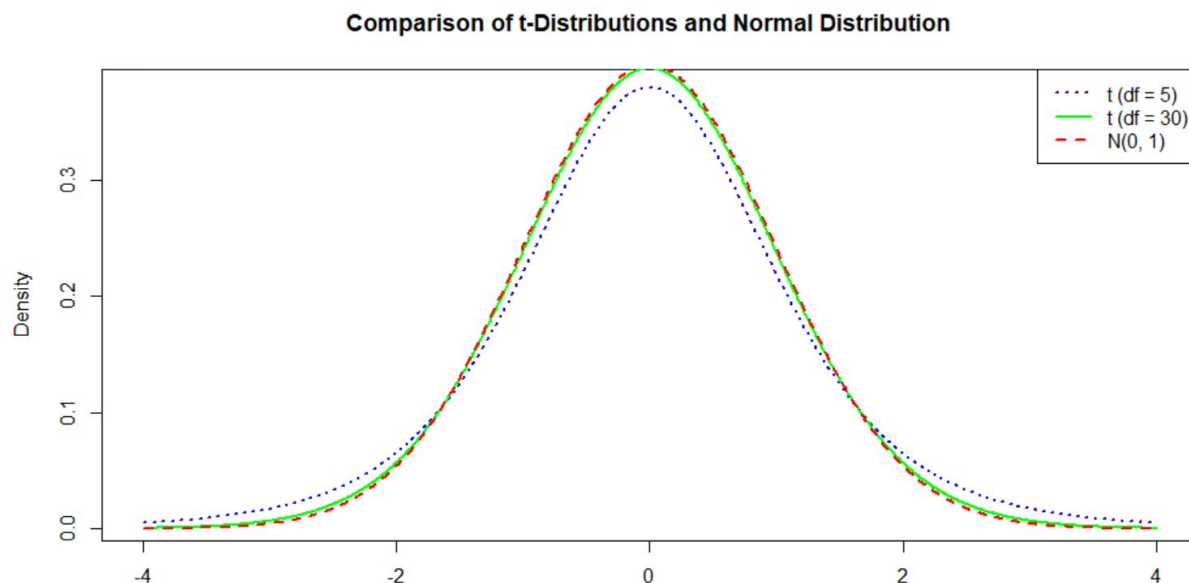


Figure 1: Comparison of t-Distributions with 5 and 30 degrees of freedom and Normal Distribution

Tests and Intervals: If we have a random sample with small sample size ($n < 30$) from a normal population with σ unknown:

Test statistic:

$$T = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

Confidence intervals:

$$\bar{x} \pm t_{\alpha/2, n-1} * \frac{s}{\sqrt{n}}$$

Example: The maker of a particular brand of smartphone claims that their battery life will allow for an average of 8 hours of talking time between recharges. I take a random sample of 6 of these phones, and measure their battery life in hours:

7.8, 8.1, 7.5, 7.7, 8.2, 7.2

Is there reason to believe that the mean battery life is less than 8 hours? Assume battery lifespans are approximately normally distributed.

Let μ be the mean battery life of this brand of phones (in hours).

Three forms of Alternative Hypothesis and P-value Determination

Example: Construct a 95% confidence interval for μ , the mean battery lifespan.

Relationship between hypothesis tests and confidence intervals:

For example, a 95% confidence interval for μ is (394.77, 401.23).

If we were to test the hypothesis $H_1 : \mu \neq 401$, since 401 is in the interval, the p-value would be greater than 0.05 and fail to reject $H_0 : \mu = 401$.

If we were to test the hypothesis $H_1 : \mu \neq 393$, since 393 is not in the interval, the p-value would be less than 0.05 and reject $H_0 : \mu = 393$.

Example: Lion-fish are native to Indo-Pacific waters but were introduced in the Atlantic in the 1990's. These spinney fish have a mean length of 13.5 inches. Recently invasive lion-fish were found 300 feet below the water's surface off the coast Florida. A random sample of 24 of these fish resulted in a mean of 16.005 inches and a standard deviation of 4.039 inches.

Is there evidence to suggest that the true mean length of these fish is different from 13.5 inches? Test using the P-value approach with $\alpha=0.01$ and assume the population is normal.

Set 23 Hypothesis Test about a population proportion

Recall that the distribution of the sample proportion, $\hat{p}$, if both np and $n(1-p)$ are at least 5, then

$$\hat{p} \sim N(\mu_{\hat{p}} = p, \sigma_{\hat{p}} = \sqrt{p(1-p)/n})$$

Using this, we can conclude:

$$Z = \frac{\hat{p} - p}{\sqrt{p(1-p)/n}}$$

Therefore, we can carry out a hypothesis test on a population proportion, (ie. $H_0 : p = p_0$ vs $H_0 : p \neq p_0$) the test statistic we will use:

$$Z = \frac{\hat{p} - p_0}{\sqrt{p_0(1-p_0)/n}}$$

where p_0 is the value proportion, p if H_0 is true.

Example: A manufacturer wants to make sure that the proportion of defective items is always no more than 3% of the total items produced. From a random sample of 500 items, the manufacturer finds 21 defective items.

Is there reason to believe the proportion of defective items is greater than 3%?

The $100(1 - \alpha)\%$ confidence interval for p : (provided there are at least 5 successes and 5 failures):

$$\hat{p} \pm z_{\alpha/2} * \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$$

Note that the formula for the confidence interval is in the same form as before:

$$\textit{estimate} \pm (c.v.) (e.s.e.)$$

Example: A survey of 2400 likely voters finds that 1379 intend to vote for a particular candidate. Find a 93% confidence interval for p , the proportion of likely voters who intend to vote for this candidate.

Sample Size Determination: Recall The $100(1-\alpha)\%$ confidence interval for p

$$\hat{p} \pm z_{\alpha/2} * \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

Let E be the **half width**, or margin of error, of our confidence interval, then

$$E = z_{\alpha/2} * \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

Solving for n :

$$n = \frac{z_{\alpha/2}^2 * \hat{p}(1-\hat{p})}{E^2}$$

Two options to estimate the sample size n with a certain confidence level:

1. $\hat{p}$ is known:

1. $\hat{p}$ is unknown:

Example: a). In the 1950's, a study in which the sample proportion of senior citizens with high blood pressure was found to be 12%. We'd like to study this proportion today. We'd like to be able to create a 90% confidence interval for the true proportion, p with a length of 0.02. How many senior citizens do we need to sample?

b). If we don't have a prior estimate for $\hat{p}$ or don't wish to use a prior estimate for $\hat{p}$ then we use:

Set 24 Comparing two population means using independent samples when population variances are known

The inferences discussed in this set concern a difference $\mu_1 - \mu_2$ between the **means of two independent populations**.

Example: We have two populations we are comparing: people with a university degree, and people without an university degree. We might want to see if the mean annual salary is different for the two populations.

Basic Assumption:

- $X_1, X_2, \dots, X_{n_1}$ is a random sample from a distribution with mean μ_1 and variance σ_1^2 .
- $Y_1, Y_2, \dots, Y_{n_2}$ is a random sample from a distribution with mean μ_2 and variance σ_2^2 .
- The X and Y samples are independent of one another.

Hypothesis tests and confidence intervals for the difference between two population means, $\mu_1 - \mu_2$, using $\bar{X} - \bar{Y}$ as an estimator. We have

- $E(\bar{X} - \bar{Y}) = \mu_1 - \mu_2$.
- $V(\bar{X} - \bar{Y}) = \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}$ and $\sigma_{\bar{X} - \bar{Y}} = \text{SD}(\bar{X} - \bar{Y}) = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$

NOTES:

Test procedures and Confidence interval for two population means with known values of σ_1 and σ_2 , two independent random samples.

We assume the Populations are Normal (near-normal) and/or n_1 and n_2 are both ≥ 30 .

Then the distribution of $\bar{X} - \bar{Y}$ is approximately normally distributed.

Hypothesis:

one tailed test	one tailed test	two tailed test
$H_0 : \mu_1 - \mu_2 \leq (\mu_1 - \mu_2)_0$	$H_0 : \mu_1 - \mu_2 \geq (\mu_1 - \mu_2)_0$	$H_0 : \mu_1 - \mu_2 = (\mu_1 - \mu_2)_0$
$H_1 : \mu_1 - \mu_2 > (\mu_1 - \mu_2)_0$	$H_1 : \mu_1 - \mu_2 < (\mu_1 - \mu_2)_0$	$H_1 : \mu_1 - \mu_2 \neq (\mu_1 - \mu_2)_0$

Test Statistic:

$$Z = \frac{(\bar{x} - \bar{y}) - (\mu_1 - \mu_2)_0}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} \sim N(0, 1)$$

Test statistic has a $N(0, 1)$ when H_0 is true.

Confidence Interval:

$$(\bar{x} - \bar{y}) \pm z_{\alpha/2} \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

Example: A study looked at the brain size of 30 autistic and 12 non-autistic boys who had received an MRI scan as toddlers. Here are the sample stats of their brain volumes in mL:

Autistic: $n_1 = 30$ $\bar{X}_1 = 1297.6$ and Non-autistic : $n_2=12$ and $\bar{X}_2 = 1179.3$

Assume the population standard deviations: $\sigma_1 = 88.4$ and $\sigma_2 = 70.7$. Is there evidence that autistic boys have larger brains on average than non-autistic boys?

Construct the 95% confidence interval for $\mu_1 - \mu_2$, the difference between the true average for brain volumes for the autistic and non-autistic boys.

Assume the brain volumes in both populations are normal and test with $\alpha=0.05$

continues...

Set 25 Comparing two population means using independent samples when population variances are UNKNOWN

Hypothesis tests and confidence intervals for the difference between two population means, $\mu_1 - \mu_2$, using $\bar{X} - \bar{Y}$ as an estimator.

We will consider three cases. Each case assumes an independent random sample is taken from each population.

Case I: Large sample size (n_1 and n_2 are both ≥ 30).

Case II: Small sample sizes from near-normal populations with **Equal variances**. Conditions:

- Small sample sizes: $n_1 < 30$ and/or $n_2 < 30$
- Populations that have normal or near-normal distributions.
- Populations have equal variances ($\sigma_1^2 \approx \sigma_2^2$).

Rule of Thumb: Assume equal variances if

$$\frac{\text{larger}(s_1, s_2)}{\text{smaller}(s_1, s_2)} \leq 1.4$$

Case III: Small sample sizes from near-normal populations with **unequal variances**. Conditions:

- Small sample sizes: $n_1 < 30$ and/or $n_2 < 30$
- Populations that have normal or near-normal distributions.
- Populations have unequal variances ($\sigma_1^2 \neq \sigma_2^2$).

Rule of Thumb: Assume unequal variances if

$$\frac{\text{larger}(s_1, s_2)}{\text{smaller}(s_1, s_2)} > 1.4$$

Hypothesis:

one tailed test	one tailed test	two tailed test
$H_0 : \mu_1 - \mu_2 \leq (\mu_1 - \mu_2)_0$	$H_0 : \mu_1 - \mu_2 \geq (\mu_1 - \mu_2)_0$	$H_0 : \mu_1 - \mu_2 = (\mu_1 - \mu_2)_0$
$H_1 : \mu_1 - \mu_2 > (\mu_1 - \mu_2)_0$	$H_1 : \mu_1 - \mu_2 < (\mu_1 - \mu_2)_0$	$H_1 : \mu_1 - \mu_2 \neq (\mu_1 - \mu_2)_0$

Case I: Large sample sizes ($n_1 \geq 30, n_2 \geq 30$) (Two-Sample Z)

Test Statistic:

$$Z = \frac{(\bar{x} - \bar{y}) - (\mu_1 - \mu_2)_0}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} \sim N(0, 1)$$

Test statistic has a $N(0, 1)$ when H_0 is true.

Confidence Interval:

$$(\bar{x} - \bar{y}) \pm z_{\alpha/2} \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

Case II: Small Samples from near-normal populations with equal variances
(Pooled Two-Sample t procedure)

Test Statistic:

$$T = \frac{(\bar{x} - \bar{y}) - (\mu_1 - \mu_2)_0}{\sqrt{s_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}} \sim t_\nu$$

Test statistic has a t distribution with $\nu = n_1 + n_2 - 2$ degrees of freedom when H_0 is true.

Pooled common sample variance:

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

Confidence Interval:

$$(\bar{x} - \bar{y}) \pm t_{\alpha/2, \nu} \sqrt{s_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}$$

Case III: Small Samples from near-normal populations with unequal variances (Unpooled Two-Sample t procedure)

Test Statistic:

$$T = \frac{(\bar{x} - \bar{y}) - (\mu_1 - \mu_2)_0}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} \sim t_\nu$$

Test statistic has a t_ν when H_0 is true.

Confidence Interval:

$$(\bar{x} - \bar{y}) \pm t_{\alpha/2, \nu} \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

where the degrees of freedom ν is the **integer part** of

$$\frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{\left(\frac{s_1^2}{n_1}\right)^2}{n_1-1} + \frac{\left(\frac{s_2^2}{n_2}\right)^2}{n_2-1}}$$

Note: always round down for degrees of freedom.

The P -value for a hypothesis test on the difference between two population means is:

Case I: Large Samples ($n_1 \geq 30, n_2 \geq 30$) (Two-Sample Z test)

one tailed test	one tailed test	two tailed test
$H_1 : \mu_1 - \mu_2 > (\mu_1 - \mu_2)_0$	$H_1 : \mu_1 - \mu_2 < (\mu_1 - \mu_2)_0$	$H_1 : \mu_1 - \mu_2 \neq (\mu_1 - \mu_2)_0$
P-value= $P(Z > z_{obs})$	P-value= $P(Z < z_{obs})$	P-value= $2 * P(Z < - z_{obs})$

Cases 2 and 3: Small Samples (Two-sample t tests)

one tailed test	one tailed test	two tailed test
$H_1 : \mu_1 - \mu_2 > (\mu_1 - \mu_2)_0$	$H_1 : \mu_1 - \mu_2 < (\mu_1 - \mu_2)_0$	$H_1 : \mu_1 - \mu_2 \neq (\mu_1 - \mu_2)_0$
P-value= $P(t_\nu > t_{obs})$	P-value= $P(t_\nu < t_{obs})$	P-value= $2 * P(t_\nu > t_{obs})$

Example 1: A sample of 100 recent A University grads and 80 recent B University grads finds that the salaries for A grads has sample mean 41000 with sample standard deviation 1000, and the salaries for B grads has sample mean 40800 with sample standard deviation 600.

Is there reason to believe the true mean salaries are different?

Construct a 95% confidence interval for $\mu_1 - \mu_2$, the difference between the two population means.

Example 2: A store conducts inventory and calculates the losses due to theft. They are considering two methods: using a security guard, installing security cameras. They try both, one at a time, each for six month periods. The monthly losses in dollars were recorded. Since the cameras are cheaper, the manager will use them unless there is enough evidence to infer that the guard was better.

Is there reason to believe the losses with guards is less than the losses with cameras?

Construct a 95% confidence interval for $\mu_1 - \mu_2$, the difference between the two population means.

Security Guard: 355, 284, 401, 398, 477, 254

Cameras: 486, 303, 270, 386, 411, 435

Example 3: Two telemarketing companies are both working for the same company. A sample of 12 telemarketers at company A have average daily sales of \$106 with standard deviation \$3. A sample of 10 telemarketers at company B have average daily sales of \$101 with standard deviation \$5.

Is there evidence to support company A's claim that it has larger sales figures than company B?

Construct a 95% confidence interval for $\mu_1 - \mu_2$, the difference between the two population means.

Set 26 Paired data

Example: Reflexes – Left versus Right Hand

For right-handed people, do they have faster reflexes with their right hands or their left hands?

Eight people were tested to see how quickly they caught a ruler dropped above their open hand. The distances the ruler fell before being caught, measured in cm, are displayed in the table below.

Subject	1	2	3	4	5	6	7	8
Left Hand	15.5	8.8	17.5	12.5	6.5	22.0	16.7	12.8
Right Hand	14.0	7.9	18.0	12.0	9.3	14.8	16.5	12.5
Difference=LH-RH	1.5	0.9	-0.5	0.50	-2.8	7.2	0.2	0.3

Note: The two measurements taken on an individual are NOT independent. Inferences on the Difference between Two Population Means: Paired Data

Scenario: Two populations with a pairing of the data between the populations. There is dependence between the elements of a pair.

- n = the number of pairs from the two random samples.
- To carry out inference on $\mu_X - \mu_Y$, we define a new random variable, the difference $D_i = X_i - Y_i$ for $i=1,2,3,\dots,n$.
- $\mu_D = \mu_X - \mu_Y$ = difference in the mean values of X and Y .
- $\bar{d}$ = the average of the random sample of differences.
- s_D = sample standard deviation of the differences.

The analysis proceeds as in the single sample case (one sample mean case) based on the differences: $D_1, D_2, \dots, D_n$

Hypothesis tests and confidence intervals for the paired data difference, μ_D , between two population means use $\bar{d}$ as an estimate with estimated standard error (e.s.e.)= $s_D/\sqrt{n}$

Paired Data: Hypothesis Tests and Confidence Interval:

one tailed test	one tailed test	two tailed test
$H_0 : \mu_D = (\mu_D)_0$	$H_0 : \mu_D = (\mu_D)_0$	$H_0 : \mu_D = (\mu_D)_0$
$H_1 : \mu_D > (\mu_D)_0$	$H_1 : \mu_D < (\mu_D)_0$	$H_1 : \mu_D \neq (\mu_D)_0$

Case I: large number of pairs ($n \geq 30$)

Test statistic:

$$Z = \frac{\bar{d} - (\mu_D)_0}{s_D/\sqrt{n}} \sim N(0, 1)$$

Confidence intervals for μ_D :

$$\bar{d} \pm z_{\alpha/2} * \frac{s_D}{\sqrt{n}}$$

Case II: small number of pairs ($n < 30$) and paired differences are distributed Normal

Test statistic:

$$T = \frac{\bar{d} - (\mu_D)_0}{s_D/\sqrt{n}} \sim t_{n-1}$$

Confidence intervals for μ_D :

$$\bar{d} \pm t_{\alpha/2, n-1} * \frac{s_D}{\sqrt{n}}$$

Example: Reflexes – Left versus Right Hand

Let μ_D = mean difference in reflexes with a ruler (LH – RH).

Calculate a 90% confidence interval for the mean difference. Assume the paired differences are normally distributed.

Example: Reflexes – Left versus Right Hand . . . Do right-handed people have a faster reflex, on average, with their right hand?

Set 27 Comparing two population proportions using large samples

Having presented methods for comparing the means of two different populations, we now turn attention to the comparison of two population proportions.

Two large sample proportions setting:

Suppose $X \sim \text{Bin}(n_1, p_1)$ and $Y \sim \text{Bin}(n_2, p_2)$ with X and Y independent variables.

Population	Proportion	Sample Sizes	Count of successes	Sample proportions
1	P_1	n_1	x_1	$\hat{p}_1$
2	P_2	n_2	x_2	$\hat{p}_2$

- Each sample is independent and samples can be of differing sizes.
- To compare the two population proportions, p_1 and p_2 , we use the difference of the sample proportions, $\hat{p}_1 - \hat{p}_2$ as a estimator of $p_1 - p_2$.
- $n_1 p_1 \geq 5$, $n_1(1 - p_1) \geq 5$ and $n_2 p_2 \geq 5$, $n_2(1 - p_2) \geq 5$
- $E(\hat{p}_1 - \hat{p}_2) = p_1 - p_2$.

- $V(\hat{p}_1 - \hat{p}_2) = \frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}$ and $SD(\hat{p}_1 - \hat{p}_2) = \sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}}$

- $\hat{p}_1 - \hat{p}_2 \sim N(p_1 - p_2, \sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}})$

Hypothesis Tests for $p_1 - p_2$:

- Hypothesis
- Test statistic
- Observed statistic
- P-value
- Conclusion as usual

A $(1 - \alpha)100\%$ CI for $p_1 - p_2$:

(Check: $n_1\hat{p}_1 \geq 5$, $n_1(1 - \hat{p}_1) \geq 5$ and $n_2\hat{p}_2 \geq 5$, $n_2(1 - \hat{p}_2) \geq 5$)

$$(\hat{p}_1 - \hat{p}_2) \pm z_{\alpha/2} * \sqrt{\frac{\hat{p}_1(1 - \hat{p}_1)}{n_1} + \frac{\hat{p}_2(1 - \hat{p}_2)}{n_2}}$$

Example: A simple random sample found that 48 out of 184 people under 30 said they snored at least once a week compared with 316 out of 811 people over 30.

Test the hypothesis that people under 30 are less likely to snore than those over 30. Approach with $\alpha=0.05$.

Find 95% CI for the true difference in the proportion of snorers in the younger and older age groups.

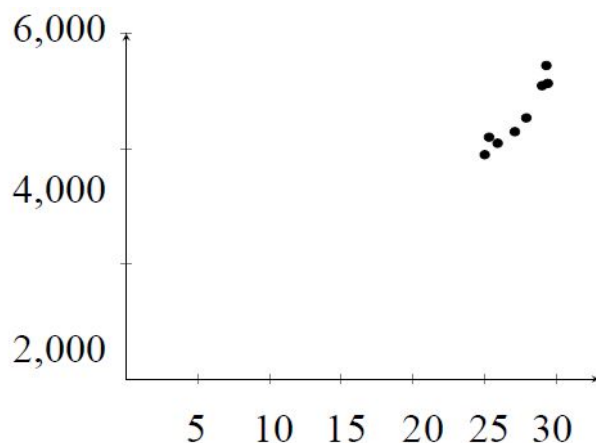
continue...

Set 28 Regression I: The simple linear Regression model

The paired data will take the form of measurements (x_1, y_1) , (x_2, y_2) , . . . , (x_n, y_n) . Over the next three sets, we will be looking at the linear (straight-line) relationship between the first and second measurements.

Example: Several of a particular species of fish are grown from eggs in tanks set at particular temperatures. After a fixed number of days, all fish are measured. We wish to examine how y , the length of the fish (in mm) depends on x , the temperature of the tank (in degrees Celsius).

tank	1	2	3	4	5	6	7	8
y	3900	4205	4100	4300	4540	5100	5450	5140
x	25.0	25.3	25.9	27.1	27.9	29.0	29.3	29.4



Looking at the plotted pairs of points, it would seem that there is a general upward trend, and that the points roughly fall along a straight line. How do we determine this numerically?

When two random variables X and Y are not independent, it is frequently of interest to assess the relationship between the two random variables.

Population Covariance:

$$\sigma_{xy} = \frac{\sum_{i=1}^N (x_i - \mu_X)(y_i - \mu_Y)}{N}$$

As we generally don't deal with population data, will not likely need to use this formula. Instead:

$$s_{xy} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{n - 1}$$

As with sample variance, we have a computational form that we will use:

$$s_{xy} = \frac{1}{n - 1} \left[\sum_{i=1}^n x_i y_i - \frac{\sum_{i=1}^n x_i \sum_{i=1}^n y_i}{n} \right]$$

Example fish example:

tank	1	2	3	4	5	6	7	8	Σ
y	3900	4205	4100	4300	4540	5100	5450	5140	36735
x	25.0	25.3	25.9	27.1	27.9	29.0	29.3	29.4	218.9
xy	97500	106386.5	106190	116530	126666	147900	159685	151116	1011974

Compute the sample covariance, s_{xy} .

Population Coefficient of Correlation:

$$\rho = \frac{\sigma_{xy}}{\sigma_x \sigma_y}$$

Sample Coefficient of Correlation:

$$r = \frac{s_{xy}}{s_x s_y}$$

Both ρ and r only take on values between -1 and 1.

Interpretation:

- $r = 0$: no linear relationship
- $r = 1$: perfect positive linear relationship
- $r = -1$: perfect negative linear relationship

Example: Calculate the coefficient of correlation for the fish data.

Population Coefficient of Determination: ρ^2

Sample Coefficient of Determination: r^2

Both ρ^2 and r^2 only take on values between 0 and 1.

Interpretation: The coefficient of determination is the proportion of the variation in y that is explained by a linear relationship with x.

Example: Calculate the coefficient of determination for the fish data.

Set 29 Regression II: Estimating model parameters

Regression Analysis: A method for analyzing data where there is one variable (called the dependent variable or response variable) whose values depend on other variables (called the independent variables or explanatory variables).

Simple linear regression model:

$$Y = \beta_0 + \beta_1 x + \varepsilon$$

- β_0 is the y-intercept of the straight line.
- β_1 is the slope of the straight line.
- ε is the difference between the observed y-value and the estimated y-value (the ones on the straight line). We sometimes will refer to as the error or the residual.

Find estimates for β_0 and β_1 (called $\hat{\beta}_0$ and $\hat{\beta}_1$ or b_0 and b_1) which will minimize the sum of the squared errors, called the **least squares estimates**.

Then regression line:

$$\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 x$$

or

$$\hat{Y} = b_0 + b_1 x$$

Formulas for our estimate:

$$b_1 = \frac{s_{xy}}{s_x^2}$$

$$b_0 = \bar{y} - b_1 \bar{x}$$

Example: A company keeps track of its annual advertising expenditure, and its annual profit. Both quantities are measured in millions of dollars. It makes sense that Profit would be explained by Advertising. So we'll have Advertising as our x values, and Profit as our y values.

	1	2	3	4	5	6	Σ
Advertising (x)	12	14	17	21	26	30	120
Profit (y)	60	70	90	100	100	120	540
xx	720	980	1530	2100	2600	3600	11530

We have, $\bar{x}=20$, $\bar{y}=90$, $s_x=7.01$ and $s_y=21.9$

Example: Suppose that 24 million dollars will be spent on advertising this year. Estimate the company's profits.

Recall that our goal was to minimize the **residuals**, or **error** (the distance from the observed values to the estimate on the line). One measure of the total error is **SSE**, the sum of the squared error terms.

$$SSE = \sum (y_i - \hat{y}_i)^2 = (n - 1)(s_y^2 - \frac{s_{xy}^2}{s_x^2})$$

The closer SSE is to zero, then the closer our points are to the regression line.

Example: Find SSE for our model.

The standard error of the estimates or residual standard error, s_ε is the estimate of the standard deviation of our error/residual terms. The greater s_ε is, then greater the variability we have in our residuals. This is another measurement of how well our line fits our points.

$$s_\varepsilon = \sqrt{\frac{SSE}{n - 2}}$$

Example: Find the residual standard error for our model.

Set 30 Regression III: Inferences about the parameters β_1 , ρ and model adequacy

In the previous set, we used the least squared method to estimate the coefficients of the linear regression model. In this set, we will present an inferential method that determines whether there is a relationship between the dependent and independent variables. For these methods to be **valid**, the following conditions on the error term, ε , must be satisfied.

- The distribution of ε is normal with mean 0, ($E(\varepsilon) = 0$).
- The standard deviation of ε is σ_ε , which is a constant regardless of the value of x .
- The error terms are all independent.

Inferences about the parameters β_1, ρ .

If a linear model is appropriate, then we would expect the slope, β_1 , to be nonzero.

Hypotheses: To test whether or not there is a linear relationship, we use

$$H_0 : \beta_1 = 0 \quad vs \quad H_1 : \beta_1 \neq 0$$

Test statistic:

$$T = \frac{b_1 - (\beta_1(0))}{s_{b_1}} \sim t_{n-2}$$

where n is the number of pairs of observations and s_{b_1} , the estimated standard error of b_1

$$s_{b_1} = \frac{s_{\varepsilon}}{\sqrt{(n-1)s_x^2}}$$

Example: For our previous example, is there evidence of a nonzero slope?

Another parameter of our population of observations is ρ , **the coefficient of correlation**. This parameter will offer us a second way of assessing the linear relationship.

Recall that When ρ is 0, there is no linear relationship between the variables.

Estimate: Our estimate for ρ is r , the sample coefficient of correlation.

$$r = \frac{s_{XY}}{s_X s_Y}$$

Hypotheses: To test whether or not there is a linear relationship, we use

$$H_0 : \rho = 0 \quad vs \quad H_1 : \rho \neq 0$$

Test Statistic:

$$T = r \sqrt{\frac{n-2}{1-r^2}} \sim t_{n-2}$$

if H_0 is true.

Example: Test whether or not our model from the previous lectures (Profit as explained by Advertising) indicates some sort of linear relationship between x and y .

Recall: $n=6$, $s_{xy} = 146$, $s_x = 7.01$ and $s_y = 21.9$.

Set 31 Goodness-of-fit test

Example: A political poll was conducted, asking a random sample of 400 residents their thoughts on a proposed increase in property tax.

	In Favour Of	Unsure	Not In Favour Of
Observed Frequency	38	137	225

Let p_1 denote the probability of being in favour of the tax,

p_2 denote the probability of being unsure,

p_3 denote the probability of being not in favour of the tax.

We wish to consider the following proportions of the population:

$p_1 = 0.1$, $p_2 = 0.35$, and $p_3 = 0.55$

A **multinomial experiment** is a generalization of binomial experiments; instead of only two outcomes, a multinomial experiment may have any (fixed) number of outcomes.

Notation: Suppose we have a multinomial experiment (n trials) with k different outcomes.

Let $f_1, f_2, \dots, f_k$ denote the observed number outcomes $1, 2, \dots, k$.

Let $p_1, p_2, \dots, p_k$ be the probability of outcome $1, 2, \dots, k$ occurring.

Note: We could also think of $p_1, \dots, p_k$ as population proportions for outcomes $1, 2, \dots, k$.

Hypothesis Test: We wish to test the claims: $p_1 = 0.1$, $p_2 = 0.35$, and $p_3 = 0.55$

Null hypothesis: H_0 : $p_1 = 0.1$, $p_2 = 0.35$, $p_3 = 0.55$

Alternative hypothesis: H_1 : at least one p_i does not equal p_{i0} .

Test Statistics:

$$\begin{aligned}\chi^2 &= \sum_{\text{all cells}} \frac{(\text{observed} - \text{expected})^2}{\text{expected}} \\ &= \sum_{i=1}^k \frac{(f_i - e_i)^2}{e_i} \sim \chi_{k-1}^2 \text{ if } H_0 \text{ is true}\end{aligned}$$

where e_i denotes the expected frequency of outcome i if the H_0 is true.

If p_{i0} is the hypothesized value of the probability for outcome i ,

$$e_i = n * p_{i0}$$

	In Favour Of	Unsure	Not In Favour Of
Observed Frequency	38	137	225
Expected Frequency			

p -value for χ^2 test is always the probability of being greater than or equal to the observed test statistic.

$$p\text{-value} = P(\chi_{k-1}^2 \geq \chi_{obs}^2)$$

Note: Provided that $e_i = n * p_{i0} \geq 5$ for all i . If $e_i < 5$ for at least one i , categories should be combined in a sensible way to correct this deficiency.

Set 32 Two-way contingency tables—testing for independence

Example: We take a random sample of 50 tuberculosis patients who were placed on a one-year antibiotic treatment. We cross-categorize them as to whether or not they had regular meetings with a doctor during the year, and whether or not they completed their treatment. Is there evidence of a relationship between regular meetings and treatment completion?

	Completed Treatment	Didn't Complete
Regular Meetings	15	4
Not Regular Meetings	10	21

Two-way contingency table: a table in which observations are categorized with respect to two different nominal variables (one variable for the rows, the other for the columns)

We may be interested in determining whether or not the two nominal variables are independent.

Hypotheses:

H_0 : the rows and columns are independent;

H_1 : the rows and columns are not independent.

Test Statistics:

$$\begin{aligned}\chi^2 &= \sum_{\text{all cells}} \frac{(\text{observed} - \text{expected})^2}{\text{expected}} \\ &= \sum_{i=1}^I \sum_{j=1}^J \frac{(f_{ij} - e_{ij})^2}{e_{ij}} \sim \chi^2_{(I-1)(J-1)} \text{ if } H_0 \text{ is true}\end{aligned}$$

where e_{ij} denotes the expected frequency of outcome ij cell if the H_0 is true.

The expected observations are, as before, the probability for a cell multiplied by the sample size, n

$$e_{ij} = \frac{(\text{ith row total})(\text{jth column total})}{n}$$

	Completed Treatment	Didn't Complete
Regular Meetings	15	4
Not Regular Meetings	10	21

p -value for χ^2 test is always the probability of being greater than or equal to the observed test statistic.

$$p\text{-value} = P(\chi^2_{(I-1)(J-1)} \geq \chi^2_{obs})$$

Note: Provided that $e_{ij} \geq 5$ for all i, j . If $e_{ij} < 5$ for at least one ij , categories should be combined in a sensible way to correct this deficiency.

Example: Much attention has recently focused on the incidence of concussions among athletes. Separate samples of soccer players, non-soccer athletes and non-athletes were selected.

	# of concussions			
	0	1	2	≥ 3
Soccer	45	25	11	10
N-S athletes	68	15	8	5
Non-athletes	45	5	3	0

Does the data suggest that # of concussions and type of individuals are independent? Carry out a test of hypotheses.

continuous...